

6-MONTH MASTER ROADMAP

AI/ML (primary) + Financial-Market Trading (secondary)
Mon 5 Oct 2026 → Sun 4 Apr 2027 · every evening 7:00-11:00 PM

182 days · 26 weeks		≈ 17.8 h AI/ML per week		≈ 7 h Trading per week		₹ 0 real capital in the plan		6 monthly checkpoints	
	MONTH 1 Foundations 5 Oct - 1 Nov 2026	MONTH 2 Core ML & Chart Reading	MONTH 3 Evaluation, Neural Nets & Risk	MONTH 4 Deep Learning, NLP & Backtesting	MONTH 5 LLMs, RAG, Agents & Python Backtesting	MONTH 6 Flagship Project & Verdict			
AI / ML >progression	- NumPy - Pandas - Matplotlib / Seaborn - Statistics & probability - Git / GitHub	- Linear algebra & gradient descent - scikit-learn pipelines - Supervised models - Clustering & PCA - Regression Project	- Metrics & imbalance - Feature engineering - NN from scratch - PyTorch basics - End-to-End ML Project	- CNNs & transfer learning - FastAPI - Docker - NLP basics - Transformers & fine-tuning	- LLM APIs & prompting - Embeddings & vector DBs - RAG pipeline + eval - RAG app (FastAPI, Docker, CI) - AI agents	- Flagship RAG + agent app - Evaluation, Docker, CI - Cloud deployment - Portfolio, resume, interviews			
TRADING >progression	- Market basics, NSE/BSE - NIFTY / SENSEX - Order types & costs - Candlesticks - Trend & support/resistance	- Volume & breakouts - Moving averages - RSI & ATR - Multi-timeframe - Setup library (30)	- Risk & position sizing - R-multiples, expectancy - Financial statements - Valuation ratios - Psychology + Strategy v1.0	88 more manual backtests - Performance statistics - Equity curve & drawdown - Paper trading begins - Edge assessment v1	- Python backtester - Robustness & out-of-sample - Walk-forward, Monte Carlo - ML-prediction autopsy - Derivatives theory only	- Momentum strategy (Python) - Paper-trading review - Strategy dossier & edge verdict - Go-live gate + policy			
MAJOR >milestones	Pandas project + statistics notebook; market journal started	Regression Project (portfolio #1); 30 documented setups	End-to-End ML Project (portfolio #2); 20 backtested trades	Image + NLP projects (#3-4); Dockerized API; 108 trades	RAG app (#5); agent mini-project; Python backtester	Flagship project deployed; go / no-go trading verdict			
MONTH-END >checkpoint	Test: Sat 31 Oct – Sun 1 Nov	Test: Sat 28 Nov – Sun 29 Nov	Test: Sat 2 Jan – Sun 3 Jan	Test: Sat 30 Jan – Sun 31 Jan	Test: Sat 6 Mar – Sun 7 Mar	Test: Sat 3 Apr – Sun 4 Apr			

HOW EVERY WEEK WORKS

Learn → Practice → Build → Review → Test. Mon-Tue learn, Wed practise, Thu build, Fri review, Sat build + backtest + weekly review, Sun closed-book test + planning.

PRIORITY ORDER

1 AI/ML → 2 College → 3 Trading → 4 Everything else. If the plan feels too heavy, shrink trading complexity — never AI/ML time. Exam week? Use Lite Mode (page 2).

TRADING GUARDRAILS

₹0 real capital. No leverage, no intraday, no F&O trading (theory only), no tips or signal groups. Real money only after the Go-Live Gate (last page) — default answer: keep paper trading.

DAILY & WEEKLY OPERATING SYSTEM

How each evening is structured · 4 hours maximum, breaks included

MONDAY - FRIDAY · AI/ML 2 h 40 · Trading 1 h · Breaks 20 min

7:00-8:30	AI/ML deep study — new concept + guided coding	90
8:30-8:40	Break — walk, water, no phone	10
8:40-9:50	AI/ML practical build — exercise, implementation or project step	70
9:50-10:00	Break	10
10:00-11:00	Trading — concept → chart/backtest → journal (last 5 min: write the log)	60

SATURDAY · build + backtest day · AI/ML 2 h 30 · Trading 1 h 10

7:00-8:20	AI/ML revision + weak-topic drills (redo missed items from scratch)	80
8:20-8:30	Break	10
8:30-9:40	AI/ML project / coding block	70
9:40-9:50	Break	10
9:50-11:00	Trading — backtesting, journal, strategy analysis; weekly review at 10:45 (fill the review box)	70

SUNDAY · lighter but productive · AI/ML 2 h · Trading 50 min · Planning 20 min

7:00-8:00	AI/ML weekly assessment (closed-book) + error analysis	60
8:00-8:10	Break	10
8:10-9:10	AI/ML light project work / refactoring	60
9:10-9:20	Break	10
9:20-10:10	Trading — journal, weekly statistics, chart study	50
10:10-10:30	Weekly scorecard + plan next week. Finish by 10:30 PM	20

Weekly load: AI/ML $\approx$ 17.8 h ($\approx$ 71%) · Trading $\approx$ 7 h ($\approx$ 29%) · breaks $\approx$ 2.3 h. Maximum 4 h in any evening; Sunday is deliberately shorter.

WEEKLY RHYTHM: Learn → Practice → Build → Review → Test

Day	AI/ML pattern	Trading pattern
Mon	LEARN — new concept + guided code	CONCEPT — rules, definitions, one worked example
Tue	LEARN → PRACTICE — exercises on the concept	CHART EXAMPLES — mark and annotate real charts
Wed	PRACTICE — implement a small version	HISTORICAL ANALYSIS — find and log past examples
Thu	BUILD — project / hands-on build step	BACKTEST / ANALYSE — test on history, record numbers
Fri	REVIEW — cheat sheet + 30-min quiz	JOURNAL + REVIEW — log, tally mistakes
Sat	BUILD + REVIEW — weak topics, push project	BACKTEST + STRATEGY + weekly review
Sun	TEST — closed-book assessment	JOURNAL + PLAN — stats, next week

RULES THAT KEEP THIS SUSTAINABLE FOR 6 MONTHS

PRIORITY ORDER — 1 AI/ML → 2 College → 3 Trading → 4 Everything else. If time runs short, cut trading complexity, never AI/ML time.

LITE MODE (exams, travel, illness, festivals) — 7:00-8:30 one AI/ML concept + one exercise, then a 15-minute trading journal; done by 9:00. At most 3 lite days in a row; never skip two days in a row.

MISSED A DAY? Don't cram. Drop that day's trading task, finish the AI/ML core task next evening, never double up.

TRADING GUARDRAILS — ₹0 real capital throughout. No leverage, no intraday, no F&O trading (theory only), no tips, Telegram or signal groups, no 'guaranteed' strategies. Real money only after the Go-Live Gate on the last page.

TRIMMED ON PURPOSE — RNN/LSTM internals, GANs, reinforcement learning, Kubernetes and deep cloud, advanced calculus, high-frequency trading, option strategies. Revisit after Month 6.

WEEKLY TIMES ON EVERY PAGE — the daily tables show what to do; the time blocks above show when.

MONTH 1 — FOUNDATIONS

5 Oct – 1 Nov 2026
Weeks 1–4 · 28 days

AI/ML OBJECTIVE

Move from a basic Python/NumPy foundation to confident data handling. Learn NumPy, Pandas and visualization properly, plus the statistics and probability that every later ML topic assumes.

TRADING OBJECTIVE

Learn how Indian markets actually work and read a basic chart: terminology, order types, costs, candlesticks, trend, support/resistance. Real capital: ₹0.

Week	Dates	AI/ML weekly goal	Trading weekly goal	Milestone
W1	5 Oct – 11 Oct	NumPy + Git/GitHub setup	Market basics & Indian market structure	<i>NumPy mini-project on GitHub</i>
W2	12 Oct – 18 Oct	Pandas I: loading, selecting, cleaning, groupby	Order types, trading costs, settlement, candlestick basics	<i>cleaned dataset + 20 candles classified</i>
W3	19 Oct – 25 Oct	Pandas II + Matplotlib/Seaborn visualization	Candlestick patterns, chart types, timeframes	<i>Project 2 (Pandas analysis) published</i>
W4	26 Oct – 1 Nov	Statistics & probability I + Month 1 checkpoint	Trends, market structure, support/resistance	<i>Month 1 checkpoint passed</i>

KEY CONCEPTS — AI/ML

NumPy arrays, indexing, broadcasting · Pandas selection, cleaning, groupby, merge · Matplotlib/Seaborn · descriptive statistics · probability · distributions & CLT · hypothesis testing · Git/GitHub

KEY CONCEPTS — TRADING

Stock, exchange, SEBI · NSE/BSE · NIFTY/SENSEX · order book & price discovery · order types · charges & T+1 settlement · candlesticks · trend · support/resistance

PRACTICAL TASKS

Daily coding drills · weekly 45-min closed-book tests · market observation journal (daily from Week 1) · chart annotation: 20 candles classified, 10 S/R levels marked, 10 breakouts studied

PROJECTS / EXERCISES

NumPy exam-score simulator · Pandas dataset analysis + visualization (published on GitHub) · 'Statistics for ML' notebook · market glossary (60+ terms)

EXPECTED OUTCOME

“I can manipulate datasets in Python and understand what I am looking at on a stock chart.”

ASSESSMENT / CHECKPOINT

Sat 31 Oct – Sun 1 Nov: NumPy/Pandas mini-task and statistics explanation; market terminology, trend/S&R and candlestick identification. Full test on the Month-1 Checkpoint page.

AI/ML focus: NumPy + Git/GitHub setup

Trading focus: Market basics & Indian market structure

Milestone: NumPy mini-project on GitHub

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
05 Oct Day 1	Mon	Set up Python env, VS Code, Jupyter; Git/GitHub basics (init, add, commit, push); NumPy array creation, dtype, shape, reshape.	What a stock, exchange, index and SEBI are; primary vs secondary market. Write a one-page 'how Indian markets work' note in your own words.	Create GitHub repo 'ai-ml-journey'; solve 10 array-creation exercises in a notebook; commit with meaningful messages.	<i>Repo live with first commit; 10 solved cells; 1-page market note.</i>
06 Oct Day 2	Tue	NumPy indexing, slicing, boolean masks, fancy indexing; the axis concept; sum/mean/std along axes.	NSE vs BSE and NIFTY 50 vs SENSEX: what each measures, how many stocks, trading hours. Note today's closing values of both.	Solve 15 indexing/masking problems (e.g., extract evens, clip outliers) before checking any solutions.	<i>15 solved problems; NSE/BSE/NIFTY/SENSEX comparison table.</i>
07 Oct Day 3	Wed	Broadcasting rules, universal functions, and why vectorization beats Python loops.	Read a stock quote: LTP, open, high, low, close, volume, bid/ask, 52-week range, circuit limits. Tabulate 5 large-cap stocks.	Time a Python loop vs vectorized code on 1M elements with %timeit; write a short explanation of the speedup.	<i>Benchmark table; 5-stock quote sheet.</i>
08 Oct Day 4	Thu	NumPy for data work: random with seeds, np.where, sorting, unique, concatenate/stack, handling NaN.	Market participants (retail, FII/DII, mutual funds, market makers) and how an order book matches buyers and sellers (price discovery).	Start mini-project: simulate 1,000 exam scores; compute mean, median, percentiles, top 10% and z-scores.	<i>Notebook with simulated scores and stats; 5-line price-discovery explanation.</i>
09 Oct Day 5	Fri	Review: build a one-page cheat sheet of 20 key NumPy functions; take a 30-min timed self-quiz.	Start the Market Observation Journal: NIFTY open/close, % change, top 3 gainers/losers, one headline, and 3 sentences on why it moved (or 'I don't know').	Finish NumPy mini-project: README, clean notebook, commit and push to GitHub.	<i>Cheat sheet; project 1a on GitHub; journal entry #1.</i>
10 Oct Day 6	Sat	Revision: redo weak NumPy exercises from scratch (broadcasting, axis ops); Python refresher on functions, comprehensions, file I/O.	Read NSE's trading-segments and indices pages; update journal; summarise NIFTY's weekly change in 5 lines. Complete the weekly review.	Implement with NumPy only: min-max and z-score normalization, moving average, Euclidean distance matrix.	<i>4 utility functions with tests; weekly review box filled.</i>
11 Oct Day 7	Sun	Weekly assessment: 45-min closed-book NumPy test (10 coding questions); mark errors and list weak topics.	Add journal entries for the week; build a 30-term glossary; preview Week 2 (order types).	Light work: refactor the Week 1 notebook into utils.py and add requirements.txt.	<i>Test score; weak-topic list; glossary v1; Week 2 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Pandas I: loading, selecting, cleaning, groupby

Trading focus: Order types, trading costs, settlement, candlestick basics

Milestone: cleaned dataset + 20 candles classified

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
12 Oct Day 8	Mon	Pandas Series and DataFrame; read_csv; head, info, describe, dtypes; shape and memory usage.	Order types: market, limit, stop-loss, SL-M, GTT. Sketch how each would fill in 3 order-book scenarios.	Load a real CSV (Titanic or IPL matches from Kaggle) and answer 10 inspection questions.	<i>Notebook with 10 answered questions; order-type sketches.</i>
13 Oct Day 9	Tue	Selection with loc/iloc, boolean filters, sorting, query(); creating new columns.	Bid-ask spread, liquidity, slippage, circuit limits. Compare spreads of a large-cap and a small-cap on a quote page.	Solve 15 filtering/sorting questions (e.g., 'all passengers under 18 in first class').	<i>15 solved queries; spread comparison note.</i>
14 Oct Day 10	Wed	Missing data: isna, fillna, dropna, imputation choices; duplicates; type conversion.	Settlement (T+1 for equities), demat basics, round-trip costs (brokerage, STT, GST, exchange fees). Cost out a ₹50,000 hypothetical delivery trade; verify current rates.	Clean a messy dataset and write a decision log justifying each cleaning choice.	<i>Clean dataset; decision log; costs worksheet.</i>
15 Oct Day 11	Thu	groupby, agg, pivot_table, value_counts, crosstab.	Candlestick anatomy: open/high/low/close, body, wicks, bullish vs bearish. Hand-draw 10 candles from OHLC rows.	Answer 10 business questions with groupby (e.g., survival rate by class and sex) and chart one.	<i>10 groupby answers; 10 hand-drawn candles.</i>
16 Oct Day 12	Fri	Review: method chaining and a Pandas cheat sheet; 20-min timed challenge on filtering plus groupby.	Journal recap; identify 5 candles on a real NIFTY daily chart and say what each shows about buyers vs sellers.	Complete a 25-problem Pandas practice set (Kaggle Learn Pandas exercises or equivalent).	<i>Cheat sheet; 25 problems done; 5 candle readings.</i>
17 Oct Day 13	Sat	Revision: redo missed Pandas problems; revisit NumPy axis operations used inside Pandas.	Classify 20 daily candles on a NIFTY 50 chart as bullish, bearish or doji; update journal; complete the weekly review.	Start Pandas Project: pick a dataset (IPL, Netflix, Spotify), write 5 questions, load and clean it.	<i>Project repo with load + clean notebook; 20 candles classified.</i>
18 Oct Day 14	Sun	Weekly assessment: 45-min test on selection, missing data and groupby (10 questions).	Journal entries; add 15 glossary terms (orders, costs, candle terms); plan Week 3.	Light work: write the project's 5 questions and expected chart types in the README.	<i>Test score; weak topics; glossary v2; Week 3 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Pandas II + Matplotlib/Seaborn visualization

Trading focus: Candlestick patterns, chart types, timeframes

Milestone: Project 2 (Pandas analysis) published

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
19 Oct Day 15	Mon	merge, join, concat; reshaping with melt and pivot; validating row counts after joins.	Candlestick patterns: doji, hammer, shooting star, engulfing; why context matters more than the pattern.	Join 2-3 related tables (e.g., matches + deliveries) and verify row counts at each step.	Merged dataset; 4 patterns described in your own words.
20 Oct Day 16	Tue	Datetime parsing, resample, rolling windows; string methods with .str.	Chart types and timeframes (daily, weekly, monthly); top-down analysis. Annotate NIFTY weekly, then daily.	Parse dates in your project dataset and build monthly or yearly trend tables.	Trend tables; 2 annotated NIFTY charts.
21 Oct Day 17	Wed	Matplotlib: figure and axes, line/bar/hist/scatter, labels, legends, subplots.	Find and mark 10 candlestick patterns on historical charts of 3 large-caps; note what the next 5 candles did (observation only).	Build 6 clear charts from your cleaned dataset with titles, labels and units.	6 chart PNGs; pattern log with 10 rows.
22 Oct Day 18	Thu	Seaborn: distributions, box plots, heatmaps, pair plots; correlation matrix; EDA storytelling.	Preview market structure: swing highs and lows. Mark them on 3 daily charts.	Write the EDA section: 8 visualizations, each with a one-line insight.	EDA notebook with 8 insights; 3 charts with swings marked.
23 Oct Day 19	Fri	Review: visualization design principles (clarity, color, no chart junk); 20-min quiz.	Journal entries; annotate 5 charts (candles, swings) and note which patterns proved unreliable.	Finalize README: problem, dataset source, methods, 5 key findings with images.	README draft; 5 annotated charts; journal updated.
24 Oct Day 20	Sat	Revision of Pandas merge/datetime/plot weak spots; peer-review your own notebook for readability.	Chart-reading drill: 10 unseen charts — trend direction, key candles, timeframe context. Complete the weekly review.	Polish and publish Project 2 (Pandas dataset analysis + visualization) on GitHub with requirements.txt.	Project 2 published; 10 chart-reading answers.
25 Oct Day 21	Sun	Weekly assessment: 45-min test on merge, datetime and plotting.	Journal entries; add 15 glossary terms (candles, timeframes); plan Week 4.	Light work: write a GitHub profile README listing your first 2 projects.	Test score; profile README; Week 4 plan.

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Statistics & probability I + Month 1 checkpoint

Trading focus: Trends, market structure, support/resistance

Milestone: Month 1 checkpoint passed

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
26 Oct Day 22	Mon	Descriptive statistics: mean, median, mode, variance, std, percentiles, skew; outliers.	Trends: uptrend (higher highs/lows), downtrend, sideways. Mark the trend on 5 charts.	Compute statistics by hand and with NumPy on a dataset; write plain-English interpretations.	<i>Stats notebook section 1; 5 trend-labelled charts.</i>
27 Oct Day 23	Tue	Probability: events, conditional probability, independence, Bayes' theorem.	Support and resistance: why levels form. Mark 10 S/R levels on historical charts and write why each matters.	Solve 10 probability problems; simulate Monty Hall and coin flips in NumPy.	<i>10 problems; simulation notebook; 10 S/R levels with reasons.</i>
28 Oct Day 24	Wed	Distributions: normal, binomial, Poisson, uniform; the Central Limit Theorem.	Trend vs range: blind-classify 10 random chart windows, then check against the full chart.	Simulate the CLT: draw 1,000 samples from a skewed distribution and plot the sample means.	<i>CLT plots; trend/range accuracy score __/10.</i>
29 Oct Day 25	Thu	Sampling, confidence intervals, hypothesis testing (p-value, t-test); correlation vs causation.	S/R flips and false breakouts: record 10 examples of broken levels and what followed.	Run a t-test with SciPy on a real dataset and explain the result in 3 plain sentences.	<i>t-test notebook; 10 breakout examples.</i>
30 Oct Day 26	Fri	Review: summarise statistics for ML on one page; 30-min quiz; prepare for tomorrow's checkpoint.	Flashcards: 40 market terms and chart concepts; journal entries.	Assemble a 'Statistics for ML' notebook with 5 worked examples and clean commentary.	<i>Stats notebook v1; 40 flashcards.</i>
31 Oct Day 27	Sat	MONTH 1 CHECKPOINT (AI/ML): timed 90-min test — NumPy/Pandas mini-task, coding problems, and explain core statistics aloud.	MONTH 1 CHECKPOINT (Trading): define 20 terms, identify trend and S/R on 3 unseen charts, read 10 candles.	Complete the tasks on the Month 1 Checkpoint page and self-score honestly.	<i>Checkpoint scores; gap list.</i>
01 Nov Day 28	Sun	Checkpoint review: redo every missed question; write a 'what I know / don't know yet' list.	Journal month summary; count charts analysed in Month 1; review mistakes.	Light work: tidy all repos; set Month 2 targets.	<i>Gap-fix list; Month 1 retrospective; Month 2 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

MONTH 1 CHECKPOINT

Sat 31 Oct - Sun 1 Nov 2026
Closed-book · timed · self-scored honestly

AI/ML TEST

1. Timed 90-min task: load a CSV with NumPy/Pandas, clean it, answer 5 questions with groupby, produce 3 clear charts. Score ____ / 10

2. Explain aloud, in plain words: mean vs median, variance, normal distribution, CLT, p-value. Score ____ / 10

3. Solve 5 coding problems without notes: indexing, broadcasting, groupby, merge, missing data. Score ____ / 10

PASS CRITERIA

Pass mark: 70% in each section (average of the three scores × 10). Below 70%: spend the next Saturday re-doing the weak topics from scratch and re-test only those items before moving on. A pass is not perfection — it means you can do the core tasks without help.

TOP 3 GAPS TO FIX NEXT MONTH (be specific)

TRADING TEST

1. Define 20 market terms (LTP, bid-ask, circuit limit, T+1, index, SL order...) in your own words. Score ____ / 10

2. Identify trend, support and resistance on 3 unseen charts. Score ____ / 10

3. Read 10 candles: what does each say about buyers vs sellers? Score ____ / 10

MONTHLY SCORECARD

AI/ML test: ____ % Trading test: ____ % Consistency: ____ / 10

Average evening hours actually studied: ____

Decision: ☐ Proceed ☐ Repeat weak topics first

MONTHLY RETROSPECTIVE — what worked · what to change · what to drop

MONTH 2 — CORE ML & CHART READING

2 Nov – 29 Nov 2026
Weeks 5–8 · 28 days

AI/ML OBJECTIVE

Learn the maths ML runs on and the complete scikit-learn workflow: preprocessing, pipelines, leakage-free evaluation, supervised and unsupervised models. Ship your first portfolio project.

TRADING OBJECTIVE

Build a technical-analysis toolkit — trendlines, volume, moving averages, RSI, ATR, multi-timeframe — and learn where each fails. Start a setup library and begin manual backtesting.

Wee k	Dates	AI/ML weekly goal	Trading weekly goal	Milestone
W5	2 Nov – 8 Nov	Linear algebra, calculus intuition, linear regression from scratch	Trendlines, volume, breakouts	'Linear Regression from Scratch' repo
W6	9 Nov – 15 Nov	scikit-learn workflow, preprocessing, leakage, pipelines	Moving averages and trend-following	Regression Project started with baseline
W7	16 Nov – 22 Nov	Supervised learning: trees, ensembles, boosting, tuning	RSI, ATR, indicator pitfalls, multi-timeframe analysis	Regression modelling complete
W8	23 Nov – 29 Nov	Unsupervised learning + Month 2 checkpoint	Setup library, manual backtest method, journal template	Month 2 checkpoint passed; Regression Project published

KEY CONCEPTS — AI/ML

Vectors, matrices, eigen-decomposition · derivatives & gradient descent · linear/logistic regression · train/test split & data leakage · pipelines · cross-validation · KNN, trees, random forests, boosting · tuning & bias-variance · K-means, DBSCAN, PCA

KEY CONCEPTS — TRADING

Trendlines & channels · volume confirmation · breakouts vs fakeouts · SMA/EMA · RSI · ATR · multi-timeframe analysis · setup-quality grading · bar-replay backtesting

PRACTICAL TASKS

Linear regression from scratch · leakage demonstration · 5-fold CV tables · model leaderboards · 30 documented setups · 15 manual practice trades · journal template

PROJECTS / EXERCISES

Linear Regression from Scratch repo · Regression Project (portfolio piece 1) · clustering/PCA notebook · setup library (30 setups)

EXPECTED OUTCOME

“I can build, tune and evaluate classic ML models with scikit-learn, and annotate a chart with trend, levels, volume and indicators — knowing where each one fails.”

ASSESSMENT / CHECKPOINT

Sat 28 – Sun 29 Nov: Regression Project review + 20-question ML test; 5 unseen charts annotated with trend, S/R, volume, MA, RSI and ATR. Full test on the Month-2 Checkpoint page.

AI/ML focus: Linear algebra, calculus intuition, linear regression from scratch

Trading focus: Trendlines, volume, breakouts

Milestone: 'Linear Regression from Scratch' repo

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
02 Nov Day 29	Mon	Vectors, dot product, norms, cosine similarity; matrices as data tables; reading shapes.	Trendlines and channels: rules for a valid line (3+ touches). Draw ascending and descending lines on 5 charts.	Implement dot product, norm and cosine similarity in NumPy; verify against np.linalg.	<i>Linear-algebra notebook part 1; 5 trendline charts.</i>
03 Nov Day 30	Tue	Matrix multiplication, transpose, inverse; why $y = Xw$ is the engine of ML; eigenvalues and a PCA preview.	Volume basics: what volume can confirm. Study 5 breakouts with and without volume support.	Compute an eigendecomposition of a covariance matrix in NumPy; plot principal directions in 2D.	<i>Eigen plot; 10 shape-debugging drills; volume notes.</i>
04 Nov Day 31	Wed	Derivatives, partial derivatives, gradients, chain rule; gradient descent intuition.	Volume + price: climax volume and dry-ups. Mark 10 volume events on charts and record outcomes.	Implement gradient descent for $y = wx + b$; plot the loss curve and compare 3 learning rates.	<i>Loss-curve plots; 10-row volume event log.</i>
05 Nov Day 32	Thu	Linear regression from first principles: MSE loss, normal equation vs gradient descent.	Breakouts vs fakeouts: collect 10 historical breakouts (5 held, 5 failed) with volume notes.	Implement linear regression from scratch on a house-price dataset; compare with the closed-form solution.	<i>From-scratch regression class; 10-row breakout table.</i>
06 Nov Day 33	Fri	Review: re-derive the MSE gradient on paper; regression metrics (MAE, RMSE, R^2); 30-min quiz.	Journal; annotate 5 charts for trend, S/R and volume in one pass; list what was ambiguous.	Add R^2 , RMSE and a residual plot to your from-scratch model; document results.	<i>Quiz score; metrics notebook; 5 annotated charts.</i>
07 Nov Day 34	Sat	Revision of weak linear-algebra items; skim an intuition-first resource (e.g., 3Blue1Brown's linear algebra series).	Setup collection: gather 10 trend-continuation setups from NIFTY 50 stocks (last 2 years) with screenshots. Complete the weekly review.	Polish the 'Linear Regression from Scratch' repo: README with the math, plots and results.	<i>Repo published; 10 setups logged.</i>
08 Nov Day 35	Sun	Weekly assessment: 45-min test on matrices, gradients and regression (10 questions).	Journal; glossary +15 (trend, volume, breakout terms); plan Week 6.	Light work: write a short note in the repo, 'How gradient descent works'.	<i>Test score; weak topics; Week 6 plan.</i>

Times — Mon-Fri: AI/ML 7:00–8:30 · break · practical 8:40–9:50 · break · trading 10:00–11:00. Sat: 7:00–8:20 · 8:30–9:40 · 9:50–11:00 (weekly review 10:45). Sun: 7:00–8:00 · 8:10–9:10 · trading 9:20–10:10 · plan 10:10–10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: scikit-learn workflow, preprocessing, leakage, pipelines

Trading focus: Moving averages and trend-following

Milestone: Regression Project started with baseline

Date	Day	AI/ML Session deep study · 7:00–8:30 (Sat 7:00–8:20)	Trading Session concept → chart → journal · 10:00–11:00 (Sat 9:50–11:00)	Practical Task AI/ML build · 8:40–9:50 (Sat 8:30–9:40)	Expected Output what exists by tonight
09 Nov Day 36	Mon	scikit-learn API (fit, predict, transform); train/test split; why a baseline comes first; data leakage.	Moving averages: SMA vs EMA and what they smooth. Plot 20/50/200 on a NIFTY chart and describe each.	Split a small dataset, train a baseline, and explain data leakage with a concrete example.	<i>Baseline notebook; 5-line leakage explanation; MA chart.</i>
10 Nov Day 37	Tue	Preprocessing: scaling (Standard, MinMax), encoding (one-hot, ordinal), imputation; fit on train only.	Moving averages as dynamic support/resistance, and their lag. Mark 10 MA bounces/breaks on charts.	Demonstrate leakage: fit a scaler on all data vs train only and compare scores.	<i>Leakage demo notebook; 10 MA examples.</i>
11 Nov Day 38	Wed	ColumnTransformer and Pipeline; saving models with joblib.	Compute SMA and EMA by hand in a spreadsheet for 20 bars; compare with the charting platform.	Build a full preprocessing + model pipeline on a mixed-type dataset and save it.	<i>Pipeline code; saved model file; MA spreadsheet.</i>
12 Nov Day 39	Thu	Linear and logistic regression in scikit-learn; reading coefficients; Ridge and Lasso regularization.	20/50 EMA crossover on 3 stocks over 2 years: count whipsaws vs sustained trends (observation only).	Predict house prices with Linear, Ridge and Lasso; compare coefficients and test error.	<i>Model comparison table; whipsaw count table.</i>
13 Nov Day 40	Fri	Review: K-fold cross-validation and why a single split is noisy; 30-min quiz.	Journal; note MA pitfalls (lag, whipsaws in ranges); annotate 5 more charts.	Run 5-fold CV on 3 models and tabulate mean ± std of the scores.	<i>CV results table; MA pitfalls note.</i>
14 Nov Day 41	Sat	Revision: pipelines and leakage; shortlist datasets for the Regression Project.	Scan 20 charts for 'pullback to 50 EMA in an uptrend' setups; record entry, stop, outcome (no execution). Complete the weekly review.	Regression Project: pick a dataset (California housing or a Bengaluru house-price set), define target and metric, build a baseline pipeline.	<i>Project repo with problem statement and baseline; 20 charts scanned.</i>
15 Nov Day 42	Sun	Weekly assessment: 45-min scikit-learn workflow test (10 questions + 1 coding drill).	Journal; glossary +15 (MA, EMA, crossover, whipsaw); plan Week 7.	Light work: add a baseline-results table to the project README.	<i>Test score; weak topics; Week 7 plan.</i>

Times — Mon–Fri: AI/ML 7:00–8:30 · break · practical 8:40–9:50 · break · trading 10:00–11:00. Sat: 7:00–8:20 · 8:30–9:40 · 9:50–11:00 (weekly review 10:45). Sun: 7:00–8:00 · 8:10–9:10 · trading 9:20–10:10 · plan 10:10–10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Supervised learning; trees, ensembles, boosting, tuning

Trading focus: RSI, ATR, indicator pitfalls, multi-timeframe analysis

Milestone: Regression modelling complete

Date	Day	AI/ML Session deep study · 7:00–8:30 (Sat 7:00–8:20)	Trading Session concept → chart → journal · 10:00–11:00 (Sat 9:50–11:00)	Practical Task AI/ML build · 8:40–9:50 (Sat 8:30–9:40)	Expected Output what exists by tonight
16 Nov Day 43	Mon	KNN and decision trees: Gini/entropy, depth control, overfitting.	RSI is a momentum oscillator; overbought/oversold is not a sell/buy signal. Compute 14-period RSI in a spreadsheet for 15 bars.	Train KNN and a decision tree; plot train vs test accuracy as k and depth vary.	<i>Over/underfit plots; RSI worksheet.</i>
17 Nov Day 44	Tue	Ensembles: bagging, random forests, feature importance, out-of-bag score.	RSI inside trends (the 40–80 zone in uptrends) and divergences. Mark 10 examples.	Compare a random forest with a single tree; inspect feature importance and OOB score.	<i>RF vs tree comparison; 10 RSI examples.</i>
18 Nov Day 45	Wed	Gradient boosting: GBM, XGBoost/LightGBM concepts; boosting vs bagging.	ATR: true range and volatility. Compute ATR(14) on 3 stocks and express it as % of price.	Train HistGradientBoosting or XGBoost and compare with random forest on identical CV folds.	<i>3-model leaderboard; ATR table.</i>
19 Nov Day 46	Thu	Hyperparameter tuning: GridSearchCV, RandomizedSearchCV; bias-variance trade-off; validation curves.	ATR-based stops: illustrate 1.5× and 2× ATR stops on 5 hypothetical entries and compare distances.	Tune RF or boosting with randomized search; plot a validation curve; report CV vs hold-out score.	<i>Tuned model; validation curve; 5 stop illustrations.</i>
20 Nov Day 47	Fri	Review: bias-variance, overfitting symptoms, when to stop tuning; 30-min quiz.	Multi-timeframe analysis: weekly trend, then daily entry. Annotate 5 charts across 2 timeframes; journal.	Regression Project: model comparison table and final model choice with reasoning.	<i>Quiz score; model-selection section; 5 two-timeframe charts.</i>
21 Nov Day 48	Sat	Revision of tree/ensemble weak spots; read the scikit-learn user guide section on ensembles.	Collect 10 more setups (pullback to MA with RSI/ATR context) and grade each A/B/C by clarity. Complete the weekly review.	Regression Project: residual analysis, error by segment, feature-importance plots.	<i>Error-analysis notebook; 20 graded setups in total.</i>
22 Nov Day 49	Sun	Weekly assessment: 45-min test on supervised learning and tuning.	Journal; glossary +15 (RSI, ATR, divergence, multi-timeframe); plan Week 8.	Light work: draft the project README's limitations section.	<i>Test score; weak topics; Week 8 plan.</i>

Times — Mon-Fri: AI/ML 7:00–8:30 · break · practical 8:40–9:50 · break · trading 10:00–11:00. Sat: 7:00–8:20 · 8:30–9:40 · 9:50–11:00 (weekly review 10:45). Sun: 7:00–8:00 · 8:10–9:10 · trading 9:20–10:10 · plan 10:10–10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/£0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Unsupervised learning + Month 2 checkpoint

Trading focus: Setup library, manual backtest method, journal template

Milestone: Month 2 checkpoint passed; Regression Project published

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
23 Nov Day 50	Mon	K-means: inertia, elbow method, silhouette score; why scaling matters for distance.	Setup library: define what qualifies as a setup; template (chart, context, trigger, stop, target, outcome).	Cluster a customer dataset, choose k with elbow and silhouette, and describe each cluster in words.	<i>Clustering notebook; setup-template sheet.</i>
24 Nov Day 51	Tue	Hierarchical clustering and DBSCAN; comparing clustering algorithms.	Add 10 setups to the library using the template, with screenshots.	Run K-means, hierarchical and DBSCAN on 'moons' and 'blobs' data; compare the results.	<i>Algorithm comparison figure; 10 setups filed.</i>
25 Nov Day 52	Wed	PCA and dimensionality reduction: explained variance; visualization with PCA/t-SNE.	Manual backtesting method: bar-replay (cover the future), write rules before looking, no hindsight edits. Test on 5 trades.	Apply PCA to a digits or wine dataset; plot in 2D; find components needed for 95% variance.	<i>PCA plots; 5 practice trades logged.</i>
26 Nov Day 53	Thu	Finalise the Regression Project: pipeline, evaluation, error analysis, README with results and limitations.	Backtest 10 trades with one simple rule (e.g., pullback to 20 EMA in an uptrend) on a single stock; record entry, stop, exit.	Push the Regression Project (portfolio piece 1) to GitHub with clear reproduction steps.	<i>Regression Project published; 10 logged trades.</i>
27 Nov Day 54	Fri	Checkpoint prep: re-read notes; do 20 mixed questions on scikit-learn, supervised and unsupervised learning.	Design the journal template: date, symbol, setup, entry, stop, target, R-result, emotion, rule followed (Y/N). Fill it with 10 trades.	Rework your 5 weakest items; code each from scratch.	<i>Corrected question set; journal template v1.</i>
28 Nov Day 55	Sat	MONTH 2 CHECKPOINT (AI/ML): review the Regression Project and take a timed 20-question test on sklearn, supervised and unsupervised learning.	MONTH 2 CHECKPOINT (Trading): on 5 unseen charts mark trend, S/R, volume, MA, RSI and ATR; explain one fakeout.	Complete the tasks on the Month 2 Checkpoint page and self-score.	<i>Checkpoint scores; gap list.</i>
29 Nov Day 56	Sun	Checkpoint review: redo every missed question; plan Month 3 (evaluation, features, first neural network).	Journal month summary: charts analysed, setups logged, mistakes; plan Month 3 (risk and fundamentals).	Light work: tidy repos; pin the Regression Project on GitHub.	<i>Gap-fix list; Month 2 retrospective; Month 3 plan.</i>

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

MONTH 2 CHECKPOINT

Sat 28 Nov – Sun 29 Nov 2026
Closed-book · timed · self-scored honestly

AI/ML TEST

1. Regression Project review: reproducible pipeline, CV results, error analysis, README that explains limitations. Score ____ / 10

2. Timed 20-question test: leakage, pipelines, bias-variance, regularization, clustering, PCA. Score ____ / 10

3. From scratch in 30 minutes: split + pipeline + cross-validation on a new dataset. Score ____ / 10

PASS CRITERIA

Pass mark: 70% in each section (average of the three scores × 10). Below 70%: spend the next Saturday re-doing the weak topics from scratch and re-test only those items before moving on. A pass is not perfection — it means you can do the core tasks without help.

TOP 3 GAPS TO FIX NEXT MONTH (be specific)

TRADING TEST

1. On 5 unseen charts mark trend, S/R, volume, MA, RSI and ATR. Score ____ / 10

2. Explain one fakeout and what could have warned you. Score ____ / 10

3. Show a setup library with 30 documented setups and a working journal template. Score ____ / 10

MONTHLY SCORECARD

AI/ML test: ____ % Trading test: ____ % Consistency: ____ / 10

Average evening hours actually studied: ____

Decision: ☐ Proceed ☐ Repeat weak topics first

MONTHLY RETROSPECTIVE — what worked · what to change · what to drop

MONTH 3 — EVALUATION, NEURAL NETS & RISK

30 Nov 2026 – 3 Jan 2027
Weeks 9-13 · 35 days

AI/ML OBJECTIVE

Learn proper model evaluation and feature engineering, finish a portfolio-grade end-to-end ML project, then build a neural network from scratch and move into PyTorch. Week 13 is a consolidation week that also absorbs exam/holiday disruption.

TRADING OBJECTIVE

Make risk the centre of your trading: position sizing, R-multiples, expectancy. Add basic fundamental analysis and trading psychology, then write and freeze Strategy v1.0.

Wee k	Dates	AI/ML weekly goal	Trading weekly goal	Milestone
W9	30 Nov – 6 Dec	Classification metrics, imbalance, classification project	Risk management I: risk per trade, position sizing, R-multiples	<i>classification baseline + position-size calculator</i>
W10	7 Dec – 13 Dec	Feature engineering, selection, model interpretation	Fundamental analysis I: financial statements	<i>feature-engineered models + 5-year company sheet</i>
W11	14 Dec – 20 Dec	End-to-end project packaging + neural-network fundamentals	Fundamental analysis II: valuation ratios	<i>End-to-End ML Project v1.0 (portfolio piece 2)</i>
W12	21 Dec – 27 Dec	PyTorch fundamentals	Trading psychology + strategy development I	<i>PyTorch MLP $\geq$ 88% + strategy rules draft</i>
W13	28 Dec – 3 Jan	Consolidation week: revision, Git polish, Month 3 checkpoint	Strategy v1.0 frozen + first 20 manual backtests	<i>Month 3 checkpoint passed; 20 backtested trades</i>

KEY CONCEPTS — AI/ML

Precision/recall/F1, ROC/PR · imbalanced data · feature engineering & selection · XGBoost/LightGBM · SHAP/permutation importance · project structure & tests · neurons, backprop · PyTorch tensors, autograd, training loop, regularization

KEY CONCEPTS — TRADING

Risk per trade · position-size formula · drawdown recovery · R-multiples · expectancy · 3 financial statements · P/E, P/B, EV/EBITDA, ROE, ROCE, D/E · behavioural biases · trading constitution · rule-based strategy

PRACTICAL TASKS

Confusion matrix by hand · NumPy NN with gradient check · Fashion-MNIST MLP $\geq$ 88% · 5-year company statement sheet · 3-company comparison · 20 backtested trades under frozen rules

PROJECTS / EXERCISES

Classification → End-to-End ML Project v1.0 (portfolio piece 2) · NumPy neural net · PyTorch MLP · position-size calculator · Strategy v1.0 document

EXPECTED OUTCOME

“I can ship an end-to-end ML project, explain backpropagation, size every trade by risk, read financial statements and follow written rules.”

ASSESSMENT / CHECKPOINT

Sat 2 – Sun 3 Jan: End-to-End project review, NumPy NN from memory, PyTorch quiz; sizing test, statements task, psychology reflection, strategy review. Full test on the Month-3 Checkpoint page.

AI/ML focus: Classification metrics, imbalance, classification project

Trading focus: Risk management I: risk per trade, position sizing, R-multiples

Milestone: classification baseline + position-size calculator

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
30 Nov Day 57	Mon	Classification metrics: confusion matrix, precision, recall, F1; the accuracy trap on imbalanced data.	Capital preservation: risk 0.5-1% of capital per trade; drawdown recovery math (a 50% loss needs a 100% gain). Build a recovery table.	Build a confusion matrix by hand for 20 predictions, then verify with scikit-learn.	<i>Hand-built vs sklearn matrix; drawdown recovery table.</i>
01 Dec Day 58	Tue	ROC-AUC vs PR-AUC; probability thresholds and business costs.	Position sizing: shares = risk amount ÷ (entry – stop). Build a calculator in a spreadsheet (₹1,00,000 paper capital).	Plot ROC and PR curves for a classifier; pick a threshold for a stated cost trade-off.	<i>Curve plots; position-size calculator.</i>
02 Dec Day 59	Wed	Imbalanced data: stratified splits, class weights, resampling cautions (SMOTE only inside CV folds).	Risk/reward and R-multiples: express outcomes in R. Compute R for 10 setups from your library.	Compare class weights vs resampling on an imbalanced dataset; show the accuracy illusion.	<i>Comparison table; 10-row R-multiple sheet.</i>
03 Dec Day 60	Thu	Classification Project kickoff: choose a dataset (Telco churn, credit default or heart disease); define problem, metric, baseline.	Expectancy = win% × avg win – loss% × avg loss. Test 2 hypothetical strategies (high win rate/low payoff vs low win rate/high payoff).	Run EDA and a logistic-regression baseline with a proper pipeline and stratified CV.	<i>Project repo; baseline metrics; expectancy worksheet.</i>
04 Dec Day 61	Fri	Review: choosing metrics by business problem; 30-min quiz.	Journal; apply sizing to 5 setups and check that no single loss exceeds your risk cap.	Error analysis on the baseline: inspect 20 misclassified rows and find patterns.	<i>Quiz score; error-pattern notes; 5 sized setups.</i>
05 Dec Day 62	Sat	Revision: metrics and thresholds; redo missed questions from scratch.	Apply position sizing and R-multiples to 10 setups using ₹1,00,000 paper capital. Complete the weekly review.	Classification Project: try 3 models; record CV precision/recall/AUC in a results table.	<i>Model results table; 10 sized setups.</i>
06 Dec Day 63	Sun	Weekly assessment: 45-min test on metrics, thresholds and imbalance.	Journal; glossary +15 (R, drawdown, expectancy, position sizing); plan Week 10.	Light work: update the project README with problem framing and metric choice.	<i>Test score; weak topics; Week 10 plan.</i>

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Feature engineering, selection, model interpretation

Trading focus: Fundamental analysis I: financial statements

Milestone: feature-engineered models + 5-year company sheet

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
07 Dec Day 64	Mon	Feature engineering: log transforms, binning, interactions, date features; target-encoding leakage risk.	Why fundamentals matter; the 3 statements at a glance (income statement, balance sheet, cash flow).	Engineer 10 new features on the churn data and measure each one's CV impact.	<i>Feature impact table; 3-statement summary sheet.</i>
08 Dec Day 65	Tue	Feature selection: correlation, mutual information, L1, permutation importance.	Income statement: revenue, EBITDA, PAT, margins. Pick one large-cap (e.g., IT or FMCG) and extract 5 years.	Compare all features vs selected subsets using CV.	<i>Selection experiment; 5-year P&L sheet.</i>
09 Dec Day 66	Wed	Model comparison with pipelines and CV; XGBoost/LightGBM for classification.	Balance sheet: assets, liabilities, equity, debt, working capital. Extract 5 years for the same company.	Build a 4-model leaderboard using identical stratified CV folds.	<i>Leaderboard table; balance-sheet sheet.</i>
10 Dec Day 67	Thu	Model interpretation: permutation importance and SHAP basics; explaining individual predictions.	Cash-flow statement: CFO, CFI, CFF; why profit ≠ cash. Extract 5 years and compare CFO with PAT.	Explain the top 5 churn/default drivers with SHAP or permutation importance in business language.	<i>Interpretation notebook; CFO vs PAT chart.</i>
11 Dec Day 68	Fri	Review: feature-engineering pitfalls (leakage, overfitting); 30-min quiz.	Journal; read the chairman's letter and MD&A of the company's annual report; note 5 insights.	Move modelling code into src/ modules (features.py, train.py, predict.py).	<i>Quiz score; modular code; 5 annual-report notes.</i>
12 Dec Day 69	Sat	Revision of weak feature-engineering topics; Kaggle-style mini-exercise on a fresh tabular dataset.	Build a 5-year 3-statement summary for your company: revenue, PAT, debt, CFO. Complete the weekly review.	Classification Project: finalize the best model, check calibration, save the pipeline.	<i>Saved pipeline; finished company sheet.</i>
13 Dec Day 70	Sun	Weekly assessment: 45-min test on feature engineering and model comparison.	Journal; glossary +15 (statements, margins, debt); plan Week 11.	Light work: draft the End-to-End ML Project README skeleton.	<i>Test score; weak topics; Week 11 plan.</i>

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: End-to-end project packaging + neural-network fundamentals

Trading focus: Fundamental analysis II: valuation ratios

Milestone: End-to-End ML Project v1.0 (portfolio piece 2)

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
14 Dec Day 71	Mon	Project structure: data/, notebooks/, src/, tests/, config; reproducibility with seeds and requirements.txt.	Valuation ratios: P/E, P/B, EV/EBITDA, dividend yield — what each means and where each misleads.	Refactor the classification project into the standard layout; add a pytest for the preprocessing step.	Clean repo layout; 1 passing test; ratio cheat sheet.
15 Dec Day 72	Tue	Neural networks: neuron, weights, activations (ReLU, sigmoid, softmax), forward pass.	Quality and leverage: ROE, ROCE, margins, debt/equity, interest coverage. Compute them for your company.	Implement the forward pass of a 2-layer network in NumPy.	NumPy forward-pass code; quality-ratio table.
16 Dec Day 73	Wed	Loss functions (MSE, cross-entropy) and backpropagation via the chain rule.	Growth: revenue and PAT CAGR. Compare 3 companies in one sector using Screener.in or annual reports.	Implement backpropagation for the 2-layer network and check gradients numerically.	Backprop code with gradient check; 3-company table.
17 Dec Day 74	Thu	Training loop from scratch: batches, learning rate, convergence; decision boundaries.	Value traps and cyclical: read one bear case for a 'cheap' stock and list 5 red flags.	Train the NumPy network on XOR or 'moons' data; plot the decision boundary.	Boundary plot; red-flag list.
18 Dec Day 75	Fri	Review: derive backprop for a small network on paper; 30-min quiz.	Journal; write a one-page 'what I would check before ever buying a stock' checklist.	Finalize the End-to-End ML Project README: results, limitations, how to run.	Quiz score; finished README; fundamentals checklist.
19 Dec Day 76	Sat	Revision: neural-network maths; re-implement the forward pass without notes.	Compare your 3 companies on valuation vs quality and write a balanced summary (not a recommendation). Complete the weekly review.	Tag End-to-End ML Project v1.0 on GitHub (portfolio piece 2); add a results screenshot.	v1.0 tag; sector comparison note.
20 Dec Day 77	Sun	Weekly assessment: 45-min test on neural nets, backprop and project structure.	Journal; glossary +15 (P/E, ROE, EV/EBITDA, CAGR); plan Week 12.	Light work: write a short post on 'what I learned building a neural net from scratch'.	Test score; weak topics; Week 12 plan.

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: PyTorch fundamentals

Trading focus: Trading psychology + strategy development I

Milestone: PyTorch MLP ≥ 88% + strategy rules draft

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
21 Dec Day 78	Mon	PyTorch tensors, autograd, CPU/GPU (Colab or Kaggle); tensor shapes and dtypes.	Biases that hurt traders: loss aversion, FOMO, overconfidence, recency. Write your own top-3 triggers.	Redo 5 NumPy exercises in PyTorch; compute a gradient with autograd and verify it by hand.	<i>PyTorch basics notebook; personal bias list.</i>
22 Dec Day 79	Tue	nn.Module, loss functions, optimizers, and the training loop (zero_grad, forward, loss, backward, step).	Process vs outcome: 'good loss' vs 'bad loss'. Create a 10-item pre-trade checklist.	Train a linear regression in PyTorch and compare it with scikit-learn.	<i>Working training loop; pre-trade checklist v1.</i>
23 Dec Day 80	Wed	Dataset and DataLoader; train an MLP on Fashion-MNIST; tracking train/val loss.	Write the Trading Constitution v1: risk limits, no-trade conditions, max open positions, daily/weekly rules.	Train an MLP on Fashion-MNIST with logging of train/val loss and accuracy.	<i>MLP baseline; Trading Constitution v1.</i>
24 Dec Day 81	Thu	Regularization and optimization: dropout, weight decay, early stopping, batch norm; SGD vs Adam.	Strategy development I: choose market (NIFTY 50/Next 50, daily, long-only delivery) and a hypothesis (e.g., trend pullback to 20/50 EMA).	Run 3 ablation experiments (dropout, weight decay, optimizer) and tabulate validation accuracy.	<i>Ablation table; hypothesis statement.</i>
25 Dec Day 82	Fri	Review: saving/loading models, PyTorch cheat sheet; 30-min quiz.	Journal; write exact rules: entry trigger, ATR-based stop, target/exit, position size, filters — one page.	Finish the MLP: reach ≥ 88% test accuracy on Fashion-MNIST; save weights and plot curves.	<i>MLP ≥ 88%; strategy rules v0.1.</i>
26 Dec Day 83	Sat	Revision: PyTorch training-loop details; write the loop from memory.	Apply rules v0.1 blindly to 10 historical charts: are they unambiguous? Fix vague words. Complete the weekly review.	Add a README and push the PyTorch MLP notebook to GitHub.	<i>Repo updated; strategy rules v0.2.</i>
27 Dec Day 84	Sun	Weekly assessment: 45-min test on PyTorch basics and regularization.	Journal; glossary +15 (bias, constitution, no-trade rules); plan the consolidation week.	Light work: review all open repos and list improvements for Week 13.	<i>Test score; weak topics; Week 13 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Consolidation week: revision, Git polish, Month 3 checkpoint

Trading focus: Strategy v1.0 frozen + first 20 manual backtests

Milestone: Month 3 checkpoint passed; 20 backtested trades

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
28 Dec Day 85	Mon	Consolidation: revisit metrics, feature engineering and NN notes; answer 20 ML interview questions aloud.	Finalize strategy rules v1.0 (entry, stop, exit, sizing, filters) and freeze them before any testing.	Fix your 3 weakest topics from weekly tests by re-coding the exercises from scratch.	Interview Q&A notes; strategy v1.0 frozen.
29 Dec Day 86	Tue	Git/GitHub polish: branches, pull requests, issues, .gitignore, good README structure.	Walk through the manual backtest protocol on 5 trades with the frozen rules; log them in the journal.	Clean all repos: README, requirements, folder layout; add a short 'what I learned' note to each.	All repos tidy; trades #1-5 logged.
30 Dec Day 87	Wed	Kaggle-style practice: an end-to-end submission on a fresh tabular dataset in 90 minutes.	Backtest trades #6-10 on NIFTY 50 stocks using the frozen rules; record R and emotions.	Submit predictions and compare with your CV score to check for leakage.	Submission; CV-vs-leaderboard note; trades #6-10.
31 Dec Day 88	Thu	Checkpoint prep: PyTorch and NN practice questions.	Backtest trades #11-15; mark any rule you had to interpret.	Rework weak items; write a PyTorch training loop from memory.	Corrected answers; trades #11-15.
01 Jan Day 89	Fri	Checkpoint prep: summarize Month 3 on one page; 30-min mixed quiz.	Backtest trades #16-20; calculate win rate and average R across all 20 trades.	Rehearse: explain your End-to-End ML Project in 3 minutes.	One-page summary; trades #16-20 and first stats.
02 Jan Day 90	Sat	MONTH 3 CHECKPOINT (AI/ML): review the End-to-End ML Project, build a NumPy NN from memory, take a timed PyTorch quiz.	MONTH 3 CHECKPOINT (Trading): sizing test, statement-reading task, psychology reflection, strategy-document review.	Complete the tasks on the Month 3 Checkpoint page and self-score.	Checkpoint scores; gap list.
03 Jan Day 91	Sun	Checkpoint review: redo missed questions; plan Month 4 (CNNs, APIs, Docker, NLP, Transformers).	Journal month summary: 20 trades, rule-following rate, top mistake; plan Month 4 (100+ trades).	Light work: refresh environments and notebooks; set Month 4 targets.	Gap-fix list; Month 3 retrospective; Month 4 plan.

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

MONTH 3 CHECKPOINT

Sat 2 Jan – Sun 3 Jan 2027
Closed-book · timed · self-scored honestly

AI/ML TEST

1. End-to-End ML Project review: README, tests, metric choice justified, model interpretation.

Score ____ / 10
2. Build a 2-layer NumPy network (forward + backprop) from memory and explain each gradient.

Score ____ / 10
3. Timed PyTorch quiz: tensors, autograd, training loop, regularization.

Score ____ / 10

TRADING TEST

1. Position-size test: 5 scenarios — compute shares, ₹ risk and R-multiple.

Score ____ / 10
2. Read an income statement, balance sheet and cash-flow statement; extract 6 key figures.

Score ____ / 10
3. Psychology reflection (top 3 triggers + countermeasures) and review of Strategy v1.0 against your Constitution.

Score ____ / 10

PASS CRITERIA

Pass mark: 70% in each section (average of the three scores × 10). Below 70%: spend the next Saturday re-doing the weak topics from scratch and re-test only those items before moving on. A pass is not perfection — it means you can do the core tasks without help.

MONTHLY SCORECARD

AI/ML test: ____ % Trading test: ____ % Consistency: ____ / 10

Average evening hours actually studied: ____

Decision: ☐ Proceed ☐ Repeat weak topics first

TOP 3 GAPS TO FIX NEXT MONTH (be specific)

MONTHLY RETROSPECTIVE — what worked · what to change · what to drop

MONTH 4 — DEEP LEARNING, NLP & BACKTESTING

4 Jan – 31 Jan 2027
Weeks 14–17 · 28 days

AI/ML OBJECTIVE

Move from classical ML to deep learning: CNNs and transfer learning, NLP and Transformers, and ship models properly with FastAPI and Docker.

TRADING OBJECTIVE

Test your strategy honestly: 88 more manual backtests (108 in total), real performance statistics, and the start of forward paper trading — all with ₹0.

Week	Dates	AI/ML weekly goal	Trading weekly goal	Milestone
W14	4 Jan – 10 Jan	CNNs and computer-vision basics	Manual backtest I: trades #21–42 + basic statistics	<i>CNN on CIFAR-10 $\geq$ 70%; 42 trades logged</i>
W15	11 Jan – 17 Jan	Transfer learning + FastAPI	Manual backtest II: trades #43–64 + performance statistics	<i>image-classification project + first ML API</i>
W16	18 Jan – 24 Jan	Docker + NLP fundamentals	Manual backtest III (#65–86) + start paper trading	<i>Dockerized ML API; paper-trading log started</i>
W17	25 Jan – 31 Jan	Transformers + NLP project + Month 4 checkpoint	Manual backtest IV (#87–108) + edge assessment v1	<i>100+ backtested trades; Month 4 checkpoint passed</i>

KEY CONCEPTS — AI/ML

Convolution, pooling, CNNs · augmentation · transfer learning · Grad-CAM · REST APIs & FastAPI · pytest · Docker · TF-IDF · embeddings · self-attention · Transformer block · Hugging Face · fine-tuning DistilBERT

KEY CONCEPTS — TRADING

Manual backtesting protocol · win rate, avg win/loss, payoff ratio, expectancy, profit factor · equity curve · max drawdown · losing streaks · regime analysis · paper-trading rules · sample-size uncertainty

PRACTICAL TASKS

CNN on CIFAR-10 $\geq$ 70% · fine-tune ResNet18 · API with 3 tests · Docker image of the ML API · DistilBERT vs TF-IDF baseline · about 22 backtested trades every week

PROJECTS / EXERCISES

Image-classification project (piece 3) · Dockerized ML API · NLP/Transformer project (piece 4) · 108-trade backtest report · paper-trading log

EXPECTED OUTCOME

“I can build CNN and NLP models, serve a model through a Dockerized API, and I have 100+ honestly logged trades with real statistics.”

ASSESSMENT / CHECKPOINT

Sat 30 – Sun 31 Jan: project reviews, run the Dockerized API from a clean folder, Transformer quiz; full 108-trade performance report and journal audit. Full test on the Month-4 Checkpoint page.

AI/ML focus: CNNs and computer-vision basics

Trading focus: Manual backtest I: trades #21-42 + basic statistics

Milestone: CNN on CIFAR-10 ≥ 70%; 42 trades logged

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
04 Jan Day 92	Mon	Image data basics; convolution, filters, stride, padding, pooling; what a feature map is.	Backtest trades #21-24 with frozen strategy v1.0; log R, emotion and rule-followed (Y/N) for each.	Apply edge and blur kernels to an image with NumPy/OpenCV; visualize feature maps.	Kernel demo notebook; trades #21-24.
05 Jan Day 93	Tue	CNN architecture in PyTorch: Conv2d, MaxPool2d, flatten, linear head; computing output shapes.	Backtest trades #25-28; tag each loss as a 'rule-following loss' or a 'rule-breaking loss'.	Build a small CNN for Fashion-MNIST; compare accuracy and parameter count with your MLP.	CNN vs MLP table; trades #25-28.
06 Jan Day 94	Wed	CIFAR-10: normalization, augmentation (flip, crop), BatchNorm; GPU training on Colab/Kaggle.	Backtest trades #29-32; compute win rate and expectancy after 32 trades.	Train a CNN on CIFAR-10 with augmentation; target ≥ 70% test accuracy.	Trained CNN; trades #29-32; first expectancy number.
07 Jan Day 95	Thu	Training diagnostics: learning curves, overfitting, learning-rate schedules.	Backtest trades #33-36; add average win, average loss and payoff ratio to the stats sheet.	Run 3 experiments (learning rate, augmentation, depth) and log results in a table.	Experiment log; trades #33-36.
08 Jan Day 96	Fri	Review: convolution maths and shape calculations; 30-min quiz.	Backtest trades #37-38; journal review: which rule gets broken most often, and why?	Visualize learned filters/feature maps; write a short analysis of what the network sees.	Quiz score; visualizations; trades #37-38.
09 Jan Day 97	Sat	Revision of CNN shapes and training diagnostics; read a short explainer on ResNet skip connections.	Backtest trades #39-42; analyze the first 42 trades by market regime. Complete the weekly review.	Polish the CNN notebook; push to GitHub with README and results.	CNN repo updated; 42 trades logged.
10 Jan Day 98	Sun	Weekly assessment: 45-min test on CNN architecture and training.	Journal; plan next week's backtests; re-read your Trading Constitution.	Light work: shortlist datasets for the image-classification project.	Test score; weak topics; Week 15 plan.

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/£0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Transfer learning + FastAPI

Trading focus: Manual backtest II: trades #43–64 + performance statistics

Milestone: image-classification project + first ML API

Date	Day	AI/ML Session deep study · 7:00–8:30 (Sat 7:00–8:20)	Trading Session concept → chart → journal · 10:00–11:00 (Sat 9:50–11:00)	Practical Task AI/ML build · 8:40–9:50 (Sat 8:30–9:40)	Expected Output what exists by tonight
11 Jan Day 99	Mon	Transfer learning: pretrained ResNet18, freezing/unfreezing layers, fine-tuning.	Backtest trades #43–46; log each trade fully before looking at the next chart.	Load a pretrained torchvision model and run inference on 10 of your own images.	<i>Inference notebook; trades #43–46.</i>
12 Jan Day 100	Tue	Fine-tune on a small dataset (flowers, plant disease or Indian food); dataset folders and transforms.	Backtest trades #47–50; calculate profit factor (gross profit ÷ gross loss) so far.	Set up the dataset and train the classification head.	<i>Dataset ready; head trained; trades #47–50.</i>
13 Jan Day 101	Wed	Evaluating vision models: confusion matrix, per-class errors, Grad-CAM.	Backtest trades #51–54; build the equity curve and max drawdown in the spreadsheet.	Error analysis: inspect 20 misclassified images and categorize the failure causes.	<i>Error notes; equity curve; trades #51–54.</i>
14 Jan Day 102	Thu	REST APIs and FastAPI: endpoints, request/response, Pydantic validation, Swagger docs.	Backtest trades #55–58; compare results across market conditions (trend vs range).	Build a /predict endpoint serving the End-to-End ML Project's model.	<i>Working local API; trades #55–58.</i>
15 Jan Day 103	Fri	Review: API testing with curl/Swagger and pytest; 30-min quiz.	Backtest trades #59–60; re-read your first 10 trades — are you applying rules the same way now?	Write 3 pytest tests for the API (valid input, invalid input, health check).	<i>3 passing tests; trades #59–60.</i>
16 Jan Day 104	Sat	Revision of transfer-learning decisions; write the project README (dataset, method, results, limitations).	Backtest trades #61–64; compute full statistics on 64 trades (win rate, avg R, expectancy, profit factor, max drawdown). Complete the weekly review.	Image-classification project: finish training, save the model, add an inference script and README.	<i>Image project v1 (portfolio piece 3); 64 trades.</i>
17 Jan Day 105	Sun	Weekly assessment: 45-min test on transfer learning, APIs and testing.	Journal; plan Week 16 (paper trading starts); re-read the paper-trading rules.	Light work: add the API to the End-to-End ML Project repo with usage notes.	<i>Test score; weak topics; Week 16 plan.</i>

Times — Mon–Fri: AI/ML 7:00–8:30 · break · practical 8:40–9:50 · break · trading 10:00–11:00. Sat: 7:00–8:20 · 8:30–9:40 · 9:50–11:00 (weekly review 10:45). Sun: 7:00–8:00 · 8:10–9:10 · trading 9:20–10:10 · plan 10:10–10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Docker + NLP fundamentals

Trading focus: Manual backtest III (#65-86) + start paper trading

Milestone: Dockerized ML API; paper-trading log started

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
18 Jan Day 106	Mon	Docker concepts: image, container, Dockerfile, layers, volumes, ports; install Docker.	Backtest trades #65-68; write paper-trading rules: signals after close, entry at next open, no hindsight edits.	Run a public container (hello-world, python) and write your first Dockerfile.	<i>Working first Dockerfile; paper-trading rules; trades #65-68.</i>
19 Jan Day 107	Tue	Containerize a FastAPI ML service; pin requirements; use environment variables.	Backtest trades #69-72; create the paper-trading log (signal date, entry, stop, target, size, status).	Build and run a Docker image for your /predict API; test with curl.	<i>Image running locally; paper log created; trades #69-72.</i>
20 Jan Day 108	Wed	NLP basics: tokenization, text cleaning, bag-of-words, TF-IDF.	Backtest trades #73-76; log any new paper-trade signals using the next-open rule.	Build a TF-IDF pipeline on a text dataset (SMS spam or movie reviews).	<i>TF-IDF pipeline; trades #73-76.</i>
21 Jan Day 109	Thu	Text classification with scikit-learn; error analysis on text; evaluation metrics.	Backtest trades #77-80; compute rolling win rate in 20-trade blocks to check stability.	Train Logistic Regression and Linear SVM on text; analyze 20 errors.	<i>Text baseline notebook; trades #77-80.</i>
22 Jan Day 110	Fri	Word embeddings: Word2Vec/GloVe intuition, similarity, limits of static embeddings; quiz.	Backtest trades #81-82; update the paper log; journal the week's biggest mistake.	Load pretrained embeddings; explore nearest neighbours and build an averaged-embedding classifier.	<i>Embedding exploration; trades #81-82.</i>
23 Jan Day 111	Sat	Revision of Docker commands and NLP baselines; push the Docker setup to GitHub.	Backtest trades #83-86; analyze by setup grade (best/worst). Complete the weekly review.	Publish the Dockerized ML API with a README (docker build, docker run, example curl).	<i>Dockerized API published; 86 trades.</i>
24 Jan Day 112	Sun	Weekly assessment: 45-min test on Docker and NLP basics.	Journal; update the paper log; plan Week 17 (finish 100+ trades).	Light work: write the NLP project goal and dataset plan for Week 17.	<i>Test score; weak topics; Week 17 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/£0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Transformers + NLP project + Month 4 checkpoint

Trading focus: Manual backtest IV (#87-108) + edge assessment v1

Milestone: 100+ backtested trades; Month 4 checkpoint passed

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
25 Jan Day 113	Mon	Attention and self-attention: queries, keys, values, softmax weights; why attention displaced RNNs.	Backtest trades #87-90; add new signals to the paper log.	Implement scaled dot-product attention in NumPy or PyTorch on a toy sequence.	Attention code; trades #87-90.
26 Jan Day 114	Tue	Transformer architecture: multi-head attention, positional encoding, encoder vs decoder, residuals.	Backtest trades #91-94; compute the distribution of losing-streak lengths so far.	Annotate a Transformer block diagram; implement a tiny self-attention layer.	Annotated diagram; mini layer; trades #91-94.
27 Jan Day 115	Wed	Hugging Face: tokenizers, pipelines, pretrained models; BERT/DistilBERT usage.	Backtest trades #95-98; analyze outcomes by sector.	Run sentiment/classification pipelines; inspect the tokenization of 10 sentences.	HF exploration notebook; trades #95-98.
28 Jan Day 116	Thu	Fine-tune DistilBERT for text classification on a Colab/Kaggle GPU; Trainer API basics.	Backtest trades #99-102; confirm you now have 100+ trades and compute full statistics.	Fine-tune on a sentiment or news-topic dataset; track accuracy and F1.	Fine-tuned model; 100+ trades; stats sheet.
29 Jan Day 117	Fri	Evaluate against the TF-IDF baseline; error analysis; model cards; Transformer quiz.	Backtest trades #103-104; edge assessment v1: is expectancy still positive after costs? With ~100 trades, win-rate uncertainty is roughly ±10 percentage points.	Write a TF-IDF vs DistilBERT comparison table covering accuracy, speed and error analysis.	Comparison table; edge assessment v1; trades #103-104.
30 Jan Day 118	Sat	MONTH 4 CHECKPOINT (AI/ML): review CNN/NLP projects, run your Dockerized API from a fresh folder, answer a Transformer concept quiz.	MONTH 4 CHECKPOINT (Trading): complete trades #105-108, produce the full performance report, review journal discipline.	Complete the tasks on the Month 4 Checkpoint page and self-score.	Checkpoint scores; 108 trades; gap list.
31 Jan Day 119	Sun	Checkpoint review: fix gaps; publish the NLP project README (portfolio piece 4).	Journal month summary; change strategy only if the data justifies it; plan Month 5.	Light work: list Month 5 goals (LLMs, RAG, Python backtesting).	Gap-fix list; Month 4 retrospective; Month 5 plan.

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

MONTH 4 CHECKPOINT

Sat 30 Jan – Sun 31 Jan 2027
Closed-book · timed · self-scored honestly

AI/ML TEST

1. Review CNN and NLP projects: README, results, honest limitations.

Score ____ / 10
2. Run your Dockerized API from a fresh folder using only the README.

Score ____ / 10
3. Concept quiz: convolution, transfer learning, attention, Transformer block, fine-tuning.

Score ____ / 10

PASS CRITERIA

Pass mark: 70% in each section (average of the three scores × 10). Below 70%: spend the next Saturday re-doing the weak topics from scratch and re-test only those items before moving on. A pass is not perfection — it means you can do the core tasks without help.

TOP 3 GAPS TO FIX NEXT MONTH (be specific)

TRADING TEST

1. Full report on 100+ trades: win rate, avg R, expectancy, profit factor, max drawdown, equity curve.

Score ____ / 10
2. Journal audit: 90%+ of entries complete; rule-adherence rate calculated.

Score ____ / 10
3. Edge assessment v1 with an honest statement of uncertainty.

Score ____ / 10

MONTHLY SCORECARD

AI/ML test: ____ % Trading test: ____ % Consistency: ____ / 10

Average evening hours actually studied: ____

Decision: ☐ Proceed ☐ Repeat weak topics first

MONTHLY RETROSPECTIVE — what worked · what to change · what to drop

MONTH 5 — LLMS, RAG, AGENTS & PYTHON BACKTESTING

1 Feb – 7 Mar 2027
Weeks 18-22 · 35 days

AI/ML OBJECTIVE

Learn how LLMs work and build with them: APIs, prompting, embeddings, vector databases, RAG and agents — with evaluation, not just demos.

TRADING OBJECTIVE

Move backtesting into Python: a vectorized backtester without look-ahead bias, robustness tests, walk-forward, Monte Carlo — and learn why naive ML price prediction fails. Derivatives are theory only.

Week	Dates	AI/ML weekly goal	Trading weekly goal	Milestone
W18	1 Feb – 7 Feb	LLM fundamentals, APIs, prompting, evaluation	Python for financial data + Pandas time series	LLM CLI app + indicators coded in Pandas
W19	8 Feb – 14 Feb	Embeddings, semantic search, vector databases	Vectorized backtester from scratch	Semantic Search project + backtester v1
W20	15 Feb – 21 Feb	RAG pipeline: build, improve, evaluate	Robustness: sensitivity, overfitting, out-of-sample	working RAG CLI + out-of-sample test
W21	22 Feb – 28 Feb	RAG app: FastAPI, UI, Docker, CI	Walk-forward, Monte Carlo, ML-prediction autopsy	RAG application published (portfolio piece 5)
W22	1 Mar – 7 Mar	AI agents + Month 5 checkpoint	Derivatives theory (understanding only, no trading)	Month 5 checkpoint passed; agent mini-project

KEY CONCEPTS — AI/ML

Tokens & context · LLM APIs · prompting & structured output · hallucination & prompt injection · embeddings · BM25 vs semantic search · FAISS/Chroma · chunking · RAG evaluation (hit@k, faithfulness) · tool calling · agent loops · guardrails

KEY CONCEPTS — TRADING

NSE data in Python · returns · look-ahead & survivorship bias · signal→position shift(1) · costs & slippage · CAGR, Sharpe, drawdown · overfitting · in/out-of-sample · walk-forward · Monte Carlo · futures, options, Greeks (theory)

PRACTICAL TASKS

LLM CLI chatbot · 20-question eval sets · semantic search over 100 FAQs · RAG with citations and hit@k · Dockerized RAG with CI · Python backtester with tests · ML-prediction autopsy

PROJECTS / EXERCISES

Semantic Search project · RAG application (portfolio piece 5) · Research Assistant Agent · Python backtester + robustness report · prediction-autopsy write-up

EXPECTED OUTCOME

“I can build and evaluate RAG apps and agents, test a strategy in Python without fooling myself, and explain why most price-prediction projects fail.”

ASSESSMENT / CHECKPOINT

Sat 6 – Sun 7 Mar: live RAG/agent demo + 20-question quiz; explain backtest, walk-forward and Monte Carlo results; derivatives quiz; backtester review. Full test on the Month-5 Checkpoint page.

AI/ML focus: LLM fundamentals, APIs, prompting, evaluation

Trading focus: Python for financial data + Pandas time series

Milestone: LLM CLI app + indicators coded in Pandas

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
01 Feb Day 120	Mon	How LLMs work: tokens, context window, next-token prediction; pretraining vs instruction tuning; temperature and top-p.	Fetch NSE stock history in Python (yfinance '.NS' tickers or CSV export); inspect OHLCV; adjusted vs unadjusted prices.	Use a tokenizer to count tokens in 5 texts; make your first LLM API call and inspect the response object.	<i>Token-count table; first API call; OHLCV frames for 5 stocks.</i>
02 Feb Day 121	Tue	LLM APIs: messages, system prompts, parameters, streaming, cost awareness; keep API keys in .env, never in Git.	Data quality: split/bonus adjustments, missing days, survivorship bias. Pandas returns (pct_change, log returns), rolling, resample.	Build a CLI chatbot with an LLM API (e.g., the Claude API, or a local/open model via Ollama or Hugging Face).	<i>Working CLI chatbot; data-quality checklist; returns computed.</i>
03 Feb Day 122	Wed	Prompting: few-shot, structured JSON output, role/task/format; basic chain-of-thought.	Plot price + moving averages (matplotlib/mplfinance); implement SMA, EMA, RSI and ATR in Pandas and match your charting platform.	Build a resume-to-JSON extraction prompt with Pydantic schema validation.	<i>Extraction script; indicator functions that match the platform.</i>
04 Feb Day 123	Thu	LLM limitations: hallucinations, outdated knowledge, prompt-injection basics; evaluating outputs.	Look-ahead bias demo: show how using today's close to trade today inflates results; fix it with shift(1).	Create a 20-question eval set; compare 2 prompts and score the outputs.	<i>Eval set + scores; look-ahead demo notebook.</i>
05 Feb Day 124	Fri	Review: LLM concepts quiz; when to use prompting vs RAG vs fine-tuning.	Journal; update paper-trade log; re-create one of your 108 manual signals with Pandas code.	Mini-project v0.1: document summarizer + Q&A CLI over a text file.	<i>Quiz score; CLI v0.1; one signal replicated in code.</i>
06 Feb Day 125	Sat	Revision of weak LLM/API concepts; read an LLM API quickstart end to end.	Code the strategy's entry/exit rules as signals in Pandas for 10 stocks; compare with 10 manual trades. Complete the weekly review.	Refine the Q&A CLI: add streaming, error handling and a README.	<i>CLI v0.2 pushed; signal-vs-manual comparison.</i>
07 Feb Day 126	Sun	Weekly assessment: 45-min test on LLM concepts and API usage.	Journal; update paper log; plan Week 19 (backtester).	Light work: list 3 candidate document sets for your RAG project.	<i>Test score; weak topics; Week 19 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Embeddings, semantic search, vector databases

Trading focus: Vectorized backtester from scratch

Milestone: Semantic Search project + backtester v1

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
08 Feb Day 127	Mon	Embeddings: vector space, cosine similarity, what embeddings capture and miss.	Signals vs positions: turn rules into a signal column; position = signal.shift(1).	Embed 20 sentences with sentence-transformers; find and visualize nearest neighbours.	<i>Embedding notebook; signal-to-position code.</i>
09 Feb Day 128	Tue	Semantic vs keyword search; BM25 vs embeddings.	Equity curve: strategy returns = position × asset returns; cumulative returns.	Build a mini semantic search over 100 FAQ items and compare with keyword search.	<i>Search demo; equity-curve plot.</i>
10 Feb Day 129	Wed	Vector databases: FAISS and Chroma; indexes, metadata filters, approximate nearest neighbour.	Add transaction costs and slippage assumptions to the backtester.	Store and query embeddings in Chroma (or FAISS) with metadata filters.	<i>Vector-store script; cost-adjusted equity curve.</i>
11 Feb Day 130	Thu	Chunking: fixed-size, overlap, semantic, structure-based; PDF parsing basics.	Performance metrics: CAGR, volatility, Sharpe (with caveats), max drawdown, win rate.	Chunk one PDF (college handbook or annual report) 3 ways and inspect retrieval quality.	<i>Chunking comparison; metrics function.</i>
12 Feb Day 131	Fri	Review: embeddings and vector-DB quiz; retrieval quality criteria.	Package backtester v1 into backtest.py with 3 unit tests; journal.	Write the Semantic Search mini-project README with demo examples.	<i>Quiz score; backtester v1 with tests; README.</i>
13 Feb Day 132	Sat	Revision of chunking and retrieval weak spots; compare two embedding models on the same data.	Run the backtester on 10 stocks, compare with buy-and-hold and with your manual trades. Complete the weekly review.	Publish the Semantic Search project on GitHub.	<i>Semantic Search repo; backtest comparison table.</i>
14 Feb Day 133	Sun	Weekly assessment: 45-min test on embeddings, vector DBs and chunking.	Journal; update paper log; plan Week 20 (robustness).	Light work: choose the RAG document set and collect the files.	<i>Test score; weak topics; Week 20 plan.</i>

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: RAG pipeline: build, improve, evaluate

Trading focus: Robustness: sensitivity, overfitting, out-of-sample

Milestone: working RAG CLI + out-of-sample test

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
15 Feb Day 134	Mon	RAG architecture: ingest, chunk, embed, store, retrieve, augment, generate.	Compare Python backtest results with manual results; investigate every discrepancy.	Draw the RAG architecture and write the ingestion script for your document set.	<i>Architecture diagram; ingestion script; discrepancy log.</i>
16 Feb Day 135	Tue	Retrieval and prompt assembly; source metadata and citations.	Parameter sensitivity: vary MA lengths and ATR multiples and make a result heatmap.	Build a retrieval function returning top-k chunks with source and page metadata.	<i>Retriever; sensitivity heatmap.</i>
17 Feb Day 136	Wed	Generation: grounded prompts, citing sources, saying 'I don't know'.	Overfitting and curve-fitting: why the 'best' parameters usually disappoint on new data.	Build an end-to-end RAG CLI over your documents with citations.	<i>RAG CLI v0.1; written overfitting note.</i>
18 Feb Day 137	Thu	Improving RAG: hybrid search, reranking, query rewriting, metadata filters.	Out-of-sample test: freeze the rules on the first ~70% of the data and test on the last ~30%.	Add hybrid search or a reranker and compare retrieval quality.	<i>Improved retriever; in-sample vs out-of-sample table.</i>
19 Feb Day 138	Fri	Evaluating RAG: hit@k, MRR, faithfulness, citation correctness; quiz.	Journal; write a one-page robustness summary: does anything survive out of sample?	Create a 25-question test set with gold source pages; compute hit@k.	<i>Eval set; hit@k numbers; robustness summary.</i>
20 Feb Day 139	Sat	Revision of RAG weak spots; error analysis on 10 bad answers (retrieval vs generation failures).	Re-run the out-of-sample test on a different stock universe; update the paper log. Complete the weekly review.	Fix the top retrieval failures; re-run the eval and record the improvement.	<i>RAG v0.2 results; cross-universe test.</i>
21 Feb Day 140	Sun	Weekly assessment: 45-min test on RAG design and evaluation.	Journal; plan Week 21; preview why ML price-prediction projects fail.	Light work: write the RAG README outline.	<i>Test score; weak topics; Week 21 plan.</i>

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AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: RAG app: FastAPI, UI, Docker, CI

Trading focus: Walk-forward, Monte Carlo, ML-prediction autopsy

Milestone: RAG application published (portfolio piece 5)

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
22 Feb Day 141	Mon	Wrap RAG in FastAPI: /ingest and /ask endpoints; streaming responses.	Walk-forward testing: rolling train/test windows; implement 3 windows.	Implement the endpoints and test them with Swagger.	<i>RAG API v0.1; walk-forward windows.</i>
23 Feb Day 142	Tue	Simple UI with Streamlit or Gradio; showing citations next to answers.	Walk-forward results: is performance consistent across windows?	Build the UI with a citations panel.	<i>RAG UI; walk-forward table.</i>
24 Feb Day 143	Wed	Dockerize the RAG app: Dockerfile, docker-compose, persistent vector-store volume, secrets via env vars.	Monte Carlo: shuffle trade results 1,000 times to see the range of possible drawdowns.	Write the Dockerfile and docker-compose; run the whole app in containers.	<i>Dockerized RAG app; drawdown distribution.</i>
25 Feb Day 144	Thu	Testing, logging and error handling; GitHub Actions CI for tests and linting.	ML-prediction autopsy I: predict tomorrow's close from past prices with a random split — it looks superb. Compare with the 'tomorrow = today' baseline.	Add pytest tests for retrieval, structured logging, and a CI workflow.	<i>CI badge; autopsy I notebook.</i>
26 Feb Day 145	Fri	Review: README with architecture diagram, demo GIF and 'how to run'.	Autopsy II: predict next-day direction with a time-based split and costs — expect near coin-flip. Explain: leakage, overfitting, non-stationarity, costs, multiple testing.	Finalize the README with architecture diagram and screenshots; tag v1.0.	<i>RAG app v1.0; autopsy write-up.</i>
27 Feb Day 146	Sat	Revision of weak spots; have a friend try the RAG app and collect feedback.	List 5 reasons your own strategy results might be misleading. Complete the weekly review.	Fix the top 3 usability issues from feedback; update docs.	<i>RAG app v1.1; honest-limits list.</i>
28 Feb Day 147	Sun	Weekly assessment: 45-min test on deployment-ready LLM apps.	Journal; update paper log; plan Week 22 (derivatives theory).	Light work: shortlist final-project ideas and pick one.	<i>Test score; weak topics; final-project idea chosen.</i>

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

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What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: AI agents + Month 5 checkpoint

Trading focus: Derivatives theory (understanding only, no trading)

Milestone: Month 5 checkpoint passed; agent mini-project

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
01 Mar Day 148	Mon	Agents: tool/function calling, ReAct loop, planning, memory.	Futures theory: contract, lot size, margin, mark-to-market, expiry; why leverage amplifies drawdowns.	Implement function calling with two tools (a calculator and a search over your RAG index).	Two-tool agent; margin/P&L worksheet.
02 Mar Day 149	Tue	Build a minimal agent loop without frameworks: model, tool call, observation, repeat, with a step limit.	Options basics: calls, puts, strike, premium, expiry; payoff diagrams.	Code the loop and log every step.	Agent loop; payoff plots for long call and put.
03 Mar Day 150	Wed	Agent frameworks overview (LangGraph or similar): when a framework helps and when it hides things.	Option pricing intuition: intrinsic vs time value, moneyness, theta decay, implied volatility.	Rebuild the same agent in a framework and compare the code.	Framework agent; premium-decay simulation.
04 Mar Day 151	Thu	Agent reliability: guardrails, tool errors, prompt injection, evaluating agents.	Greeks overview (delta, gamma, theta, vega) and why SEBI's own studies find most individual F&O traders lose money.	Add input validation, a step limit, and 10 adversarial test cases.	Guardrailed agent; adversarial tests; Greeks cheat sheet.
05 Mar Day 152	Fri	Review: agents quiz; design a research-assistant agent (RAG + calculator + file tool).	Write 'Why I am not trading derivatives yet': list the conditions that must be true first.	Build the Research Assistant Agent mini-project and README.	Quiz score; agent mini-project; derivatives memo.
06 Mar Day 153	Sat	MONTH 5 CHECKPOINT (AI/ML): live demo of the RAG app and agent; answer a 20-question LLM/RAG/agent quiz.	MONTH 5 CHECKPOINT (Trading): explain backtest stats and robustness results; derivatives-theory quiz; review the Python backtester.	Complete the tasks on the Month 5 Checkpoint page and self-score.	Checkpoint scores; gap list.
07 Mar Day 154	Sun	Checkpoint review: fix gaps; freeze the final-project spec (scope, MVP features, success metrics).	Journal month summary; update paper-trading stats; plan Month 6.	Light work: create the final-project repo and project board.	Gap-fix list; Month 5 retrospective; final-project spec.

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

MONTH 5 CHECKPOINT

Sat 6 Mar – Sun 7 Mar 2027
Closed-book · timed · self-scored honestly

AI/ML TEST

1. Live demo of the RAG app (citations, eval numbers) and the agent mini-project.

Score ____ / 10
2. 20-question quiz: embeddings, chunking, vector DBs, RAG evaluation, tool calling, prompt injection.

Score ____ / 10
3. Explain one failure case of your RAG and how you fixed it.

Score ____ / 10

TRADING TEST

1. Explain backtest, walk-forward and Monte Carlo results — and what they do NOT prove.

Score ____ / 10
2. Explain why naive ML price prediction fails: leakage, overfitting, non-stationarity, costs.

Score ____ / 10
3. Derivatives theory quiz (futures margin, option payoff, theta, implied volatility) + backtester code review.

Score ____ / 10

PASS CRITERIA

Pass mark: 70% in each section (average of the three scores × 10). Below 70%: spend the next Saturday re-doing the weak topics from scratch and re-test only those items before moving on. A pass is not perfection — it means you can do the core tasks without help.

MONTHLY SCORECARD

AI/ML test: ____ % Trading test: ____ % Consistency: ____ / 10

Average evening hours actually studied: ____

Decision: ☐ Proceed ☐ Repeat weak topics first

TOP 3 GAPS TO FIX NEXT MONTH (be specific)

MONTHLY RETROSPECTIVE — what worked · what to change · what to drop

MONTH 6 — FLAGSHIP PROJECT & VERDICT

8 Mar – 4 Apr 2027
Weeks 23–26 · 28 days

AI/ML OBJECTIVE

Combine everything into one portfolio-quality flagship project: an Annual-Report Analyst (RAG + agent tools) with evaluation, Docker, CI and a live cloud deployment — then polish your GitHub, resume and interview skills.

TRADING OBJECTIVE

Add a basic quantitative strategy (momentum), review paper trading honestly, write the strategy dossier and run the go-live gate. The default verdict: keep paper trading.

Week	Dates	AI/ML weekly goal	Trading weekly goal	Milestone
W23	8 Mar – 14 Mar	Final project: design, ingestion, baseline RAG; cloud basics	Quant concepts: momentum strategy + portfolio thinking	<i>final-project baseline + momentum backtest</i>
W24	15 Mar – 21 Mar	Final project core build: agent tools, API, UI	Paper-trading review & performance analysis	<i>final-project demo v0.5</i>
W25	22 Mar – 28 Mar	Final project: evaluation, testing, Docker, CI, cloud deploy	Strategy dossier, edge assessment, go-live gate	<i>final project deployed</i>
W26	29 Mar – 4 Apr	Portfolio polish + final checkpoint	Final trading report + trading policy	<i>final project v1.0; final checkpoint passed</i>

KEY CONCEPTS — AI/ML

Project design & scoping · PDF parsing · metadata-aware chunking · grounded answers with page citations · agent tools & guardrails · FastAPI + UI · evaluation harness & ablations · Docker, GitHub Actions · cloud basics & free-tier deployment · README, demo, portfolio

PRACTICAL TASKS

Ingest 5–8 annual reports · 50-question eval set · 4 ablations · cloud deployment · demo video · mock interviews · momentum backtest with costs

EXPECTED OUTCOME

“I have a deployed, documented flagship AI project, a portfolio a recruiter can read, and an evidence-based verdict on my trading strategy.”

KEY CONCEPTS — TRADING

Quant research workflow · momentum factor · correlation & diversification · paper-vs-backtest comparison · slippage audit · rule-adherence audit · mistake taxonomy · edge assessment · risk of ruin · go-live gate · trading policy

PROJECTS / EXERCISES

FLAGSHIP: Annual-Report Analyst (RAG + agent, deployed) — swap the domain if you prefer (campus handbook, government schemes, research papers) · polished GitHub profile · resume v1 · strategy dossier · Trading Policy

ASSESSMENT / CHECKPOINT

Sat 3 – Sun 4 Apr: end-to-end project presentation + 90-minute rebuild; strategy defence and go-live gate. Full test on the Month-6 Checkpoint page.

AI/ML focus: Final project: design, ingestion, baseline RAG; cloud basics

Trading focus: Quant concepts: momentum strategy + portfolio thinking

Milestone: final-project baseline + momentum backtest

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
08 Mar Day 155	Mon	Final project design: problem, users, MVP features, architecture, evaluation plan, 4-week milestones.	Quant research workflow: hypothesis, data, test, robustness, costs, forward test; investing vs trading vs quant.	Write DESIGN.md and open a GitHub Issue for each milestone.	DESIGN.md; project board; one-page quant workflow.
09 Mar Day 156	Tue	Data collection and parsing: download 5-8 annual-report PDFs for 3-4 companies; PyMuPDF/pdfplumber; table-extraction pitfalls.	Momentum and mean reversion as factors; define a simple 12-1 month momentum rule on NIFTY 50 stocks.	Write the PDF ingestion script and extract text with page numbers.	Ingestion script; raw text store; momentum rule spec.
10 Mar Day 157	Wed	Chunking with metadata (company, year, section, page); building the vector index.	Backtest momentum with monthly rebalancing (equal-weight) in Pandas, including costs.	Build the index with metadata filters; test 10 retrieval queries.	Vector index; momentum backtest v1.
11 Mar Day 158	Thu	Baseline RAG: retrieval plus grounded answers with page citations.	Evaluate momentum: CAGR, max drawdown, turnover, performance by year.	Run a 30-question baseline evaluation and record hit@k and answer correctness.	Baseline CLI; eval v0; momentum stats.
12 Mar Day 159	Fri	Cloud fundamentals: VM vs container service vs serverless; regions, costs, free tiers; secrets management.	Correlation and diversification: correlation of 5 stocks; portfolio vs single-stock volatility; compare momentum with your swing strategy.	Deploy a tiny FastAPI 'hello' service on a free-tier platform (e.g., Hugging Face Spaces, Render or Railway; check current limits).	Live hello URL; correlation matrix; comparison note.
13 Mar Day 160	Sat	Revision of cloud/deploy concepts; refine chunking based on baseline failures.	Review momentum results for overfitting risk; test a different lookback to check stability. Complete the weekly review.	Final project: improve ingestion (strip headers/footers, detect sections) and re-run the eval.	Eval v1; sensitivity table.
14 Mar Day 161	Sun	Weekly assessment: 45-min test on cloud basics and RAG design decisions.	Journal; update paper log; plan Week 24.	Light work: update DESIGN.md with what changed and why.	Test score; weak topics; Week 24 plan.

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Final project core build: agent tools, API, UI

Trading focus: Paper-trading review & performance analysis

Milestone: final-project demo v0.5

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15 Mar Day 162	Mon	Agent tools: financial-ratio calculator (ROE, margins, debt/equity) using numbers retrieved from reports; company-compare tool.	Compile paper-trading stats: number of trades, win rate, average R, expectancy, max drawdown.	Implement the tools with input validation and unit tests.	<i>Tool module + tests; paper-trading stats sheet.</i>
16 Mar Day 163	Tue	Agent routing and guardrails: when to retrieve vs compute; refuse investment advice and price predictions.	Compare paper vs backtest expectancy; look for differences in fills (the next-open assumption).	Implement the agent loop with step limit, citations and refusal policy.	<i>Agent v0.1; paper-vs-backtest comparison.</i>
17 Mar Day 164	Wed	FastAPI backend: /ask, /compare, session memory; Pydantic schemas; error handling.	Slippage audit: actual next-open vs assumed entry; estimate the average cost drag.	Build the backend and test with Swagger and pytest.	<i>Backend v0.1; slippage numbers.</i>
18 Mar Day 165	Thu	Streamlit/Gradio front end with a citations panel and company selector.	Rule-adherence audit: what share of paper trades followed the rules exactly? List every deviation.	Build the UI and connect it to the backend.	<i>UI v0.1; adherence %.</i>
19 Mar Day 166	Fri	Milestone review: demo v0.5 to a friend; capture 10 feedback items; prioritize the top 3.	Mistake taxonomy: classify every mistake (rule break, oversizing, early exit, hesitation); journal.	Fix the top 3 issues and re-test end to end.	<i>Demo v0.5; feedback list; mistake taxonomy.</i>
20 Mar Day 167	Sat	Revision of agent/RAG weak spots; read one open-source RAG-evaluation write-up.	Review all paper trades against the rules; update stats; decide whether to continue paper trading unchanged. Complete the weekly review.	Add logging and basic rate limiting; push all changes.	<i>Project v0.6; paper-review notes.</i>
21 Mar Day 168	Sun	Weekly assessment: 45-min test on agent design and API design.	Journal; plan Week 25 (dossier and go-live gate).	Light work: prepare the 50-question evaluation set for Week 25.	<i>Test score; weak topics; Week 25 plan.</i>

Times — Mon-Fri: AI/ML 7:00-8:30 · break · practical 8:40-9:50 · break · trading 10:00-11:00. Sat: 7:00-8:20 · 8:30-9:40 · 9:50-11:00 (weekly review 10:45). Sun: 7:00-8:00 · 8:10-9:10 · trading 9:20-10:10 · plan 10:10-10:30. Exam or holiday day → Lite Mode (page 2). All trading is paper/₹0.

WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Final project: evaluation, testing, Docker, CI, cloud deploy

Trading focus: Strategy dossier, edge assessment, go-live gate

Milestone: final project deployed

Date	Day	AI/ML Session deep study · 7:00–8:30 (Sat 7:00–8:20)	Trading Session concept → chart → journal · 10:00–11:00 (Sat 9:50–11:00)	Practical Task AI/ML build · 8:40–9:50 (Sat 8:30–9:40)	Expected Output what exists by tonight
22 Mar Day 169	Mon	Evaluation harness: 50-question test set; hit@k, citation accuracy, answer correctness (manual 1–5, optional LLM judge used with caution).	Strategy dossier: hypothesis, rules, data, costs, stats, limitations in one clear document.	Build the eval script and run the first full evaluation.	<i>Eval harness; eval v2 baseline; dossier outline.</i>
23 Mar Day 170	Tue	Ablations: chunk size, top-k, reranker, prompt variants; record a results table.	Edge assessment: expectancy with uncertainty, sample size, out-of-sample and walk-forward results, paper results; reach a verdict.	Run 4 ablations and pick the best configuration with evidence.	<i>Ablation table; edge-assessment verdict.</i>
24 Mar Day 171	Wed	Fix the top failure modes found in evaluation; add 'cannot find in documents' handling.	Go-live gate checklist: the criteria that must be met before real money is even considered.	Fix issues and re-run the evaluation; record the improvement.	<i>Eval v3; written go-live gate.</i>
25 Mar Day 172	Thu	Dockerize the full project; docker-compose; GitHub Actions CI (tests, lint, build).	Risk of ruin and drawdown tolerance: how large a drawdown could you really tolerate? Simulate with your trade distribution.	Write Docker and CI files; run a clean build.	<i>Dockerized project; green CI; drawdown-tolerance note.</i>
26 Mar Day 173	Fri	Deploy to a free-tier cloud service; smoke tests; basic cost and abuse protection.	Plan the next 6 months of trading learning: more paper trading, new data, what to improve.	Deploy and test the live URL end to end.	<i>Live URL; next-6-month trading plan.</i>
27 Mar Day 174	Sat	Revision and buffer: fix deployment issues; write the limitations section.	Final strategy verdict draft: keep paper trading, revise, or retire — justified with numbers. Complete the weekly review.	Final project: security review (no secrets in repo, input validation) and add a LICENSE.	<i>Stable deployment; verdict draft.</i>
28 Mar Day 175	Sun	Weekly assessment: 45-min mock technical interview on your final project.	Journal; plan the final week.	Light work: collect screenshots and numbers for the README.	<i>Mock-interview notes; Week 26 plan.</i>

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
What did I learn?	What did I learn?	AI/ML: ____ / 10
What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

AI/ML focus: Portfolio polish + final checkpoint

Trading focus: Final trading report + trading policy

Milestone: final project v1.0; final checkpoint passed

Date	Day	AI/ML Session deep study · 7:00-8:30 (Sat 7:00-8:20)	Trading Session concept → chart → journal · 10:00-11:00 (Sat 9:50-11:00)	Practical Task AI/ML build · 8:40-9:50 (Sat 8:30-9:40)	Expected Output what exists by tonight
29 Mar Day 176	Mon	Polish: README with problem, architecture diagram, eval results, limitations, demo GIF.	Final performance report: 108+ backtested trades, Python backtest, walk-forward, paper trades.	Finalize docs and add a usage section.	README v1; trading performance report.
30 Mar Day 177	Tue	Demo video (2-3 min) and a short write-up (blog or LinkedIn) explaining design decisions.	Six-month journal review: top 5 recurring mistakes and best habits.	Record and edit the demo; publish the post.	Demo video; write-up; journal review.
31 Mar Day 178	Wed	Portfolio: pin 4-6 best repos, profile README, clean notebooks, consistent naming.	Derivatives theory refresher: answer 10 questions without notes (checking understanding, not intent to trade).	Polish each repo's README top section: what, why, results, how to run.	GitHub profile ready; derivatives refresher score.
01 Apr Day 179	Thu	Resume bullets and mock interviews: ML basics, project walk-through, RAG system design.	Write a one-page Trading Policy: continue paper trading, criteria for any future real capital, risk limits.	Update your resume with measurable project outcomes; run 2 mock interviews.	Resume v1; Trading Policy.
02 Apr Day 180	Fri	Final code review and cleanup; tag v1.0 releases for the final project and top repos.	Final consistency review: rule adherence % across 6 months; journal completeness.	Run the release checklist: tests green, README accurate, no secrets, links work.	v1.0 tags; release checklist done.
03 Apr Day 181	Sat	FINAL CHECKPOINT (AI/ML): present the final project end to end; rebuild a small RAG/agent feature from memory in 90 minutes.	FINAL CHECKPOINT (Trading): defend your strategy, stats and limits; run the go-live gate honestly.	Complete the tasks on the Month 6 Checkpoint page and self-score.	Final scores; go/no-go verdict.
04 Apr Day 182	Sun	Retrospective: what to learn next (MLOps, advanced LLM evaluation, fine-tuning); plan the next 6 months.	Plan the next 6 months of trading learning; keep the ₹0-capital default unless every gate is met.	Light work: celebrate; share the project with 3 people for feedback.	6-month retrospective; next-6-month plan.

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WEEKLY REVIEW — complete on Saturday, finish scoring on Sunday

AI/ML REVIEW	TRADING REVIEW	WEEKLY SCORE
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What can I code without help?	Charts analysed: ____ Hypothetical trades recorded: ____	Trading: ____ / 10
Which concepts remain weak?	Mistakes I made:	Consistency: ____ / 10
What did I build?	Did I follow my rules? ☐ Yes ☐ Partly ☐ No	#1 fix for next week:

FINAL CHECKPOINT

Sat 3 Apr – Sun 4 Apr 2027
Closed-book · timed · self-scored honestly

AI/ML TEST

1. Present the final project end to end (8 min): problem, architecture, evaluation, deployment, limitations.

Score ____ / 10
2. Rebuild a small RAG/agent feature from memory in 90 minutes.

Score ____ / 10
3. Portfolio audit: 4+ polished repos, README quality, working links, no secrets committed.

Score ____ / 10

TRADING TEST

1. Defend your strategy: rules, statistics, limits, and what would invalidate it.

Score ____ / 10
2. Run the go-live gate checklist honestly (every item must be true).

Score ____ / 10
3. Deliver the Trading Policy: risk limits, continue-paper rule, criteria for any future capital.

Score ____ / 10

PASS CRITERIA

Final verdict: AI/ML — can you rebuild and explain the flagship project unaided? Trading — does every go-live gate item hold? If any gate item fails, the answer is simply: keep paper trading and keep the ₹0 default.

MONTHLY SCORECARD

AI/ML test: ____ % Trading test: ____ % Consistency: ____ / 10

Average evening hours actually studied: ____

Decision: ☐ Proceed ☐ Repeat weak topics first

TOP 3 GAPS TO FIX NEXT MONTH (be specific)

MONTHLY RETROSPECTIVE — what worked · what to change · what to drop

AI / ML — OUTCOME CHECKLIST

- ☐ Strong Python / NumPy / Pandas / visualization foundation
- ☐ Statistics, probability and linear algebra needed for ML
- ☐ ML fundamentals: scikit-learn, evaluation, feature engineering
- ☐ Multiple projects: Regression, End-to-End ML, Image, NLP, RAG, Agent
- ☐ Deep-learning fundamentals: PyTorch, CNNs, Transformers
- ☐ LLM / embeddings / vector DB / RAG / agent understanding
- ☐ GitHub portfolio with clear READMEs (4–6 pinned repos)
- ☐ One flagship project: deployed, Dockerized, CI, evaluated
- ☐ Deployment fundamentals: APIs, Docker, cloud basics

TRADING — OUTCOME CHECKLIST

- ☐ Market fundamentals and Indian market structure (NSE/BSE, NIFTY/SENSEX)
- ☐ Technical-analysis foundation: trend, S/R, volume, MA, RSI, ATR
- ☐ Basic fundamental analysis: statements and valuation ratios
- ☐ Risk management: sizing, R-multiples, drawdown, risk of ruin
- ☐ One clearly defined strategy with written rules (v1.0, frozen)
- ☐ 100+ backtested trades (108 planned) and a trading journal
- ☐ Python backtester, walk-forward and Monte Carlo analysis
- ☐ Paper-trading experience and performance statistics
- ☐ An honest verdict on whether the strategy has an edge
- ☐ Derivatives understood in theory — not traded

GO-LIVE GATE — ALL must be true

1. 100+ backtested trades with positive expectancy after costs.
2. Out-of-sample and walk-forward results do not collapse.
3. 30+ forward paper trades; expectancy broadly consistent with the backtest.
4. Rule adherence of 90%+ for 12 consecutive weeks.
5. The maximum drawdown seen in testing is one you could truly tolerate.
6. Emergency savings intact; any capital is money you can afford to lose entirely.
7. Delivery only, no leverage, tiny size, and reviewed first with a trusted adult or qualified adviser.

HOW TO READ THE GATE

The gate cannot realistically be passed before mid-April 2027 (paper trading starts in Week 16, and 12 consecutive weeks of rule adherence are required), so the plan's built-in answer at Month 6 is: **keep paper trading with ₹0**. Passing the gate means real money becomes *something you may consider*, not something you must do.

FREE RESOURCES — AI/ML

NumPy/Pandas/scikit-learn official docs · Kaggle Learn · 3Blue1Brown (linear algebra, neural nets) · StatQuest · Andrej Karpathy 'Neural Networks: Zero to Hero' · PyTorch tutorials · Hugging Face course · fast.ai Practical Deep Learning

FREE RESOURCES — TRADING

Zerodha Varsity (free, India-focused) · NSE, BSE and SEBI investor-education pages · Screener.in · TradingView · company annual reports (investor-relations pages) · 'Trading in the Zone' (psychology) · later: 'Advances in Financial Machine Learning' (why finance ML is hard)

READ THIS

Important: This is an educational study plan, not investment advice. Markets involve risk of loss; most individual traders lose money, and SEBI's own studies have found that the large majority of individual F&O traders lose. Verify current exchange rules, charges and holiday calendars on nseindia.com and bseindia.com, and check current free-tier limits of any cloud or API service before relying on it. Never share API keys or commit them to Git.